

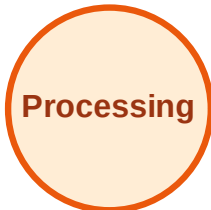
BFS Traversal

Step 1: Start at A

Legend



Complete



Processing



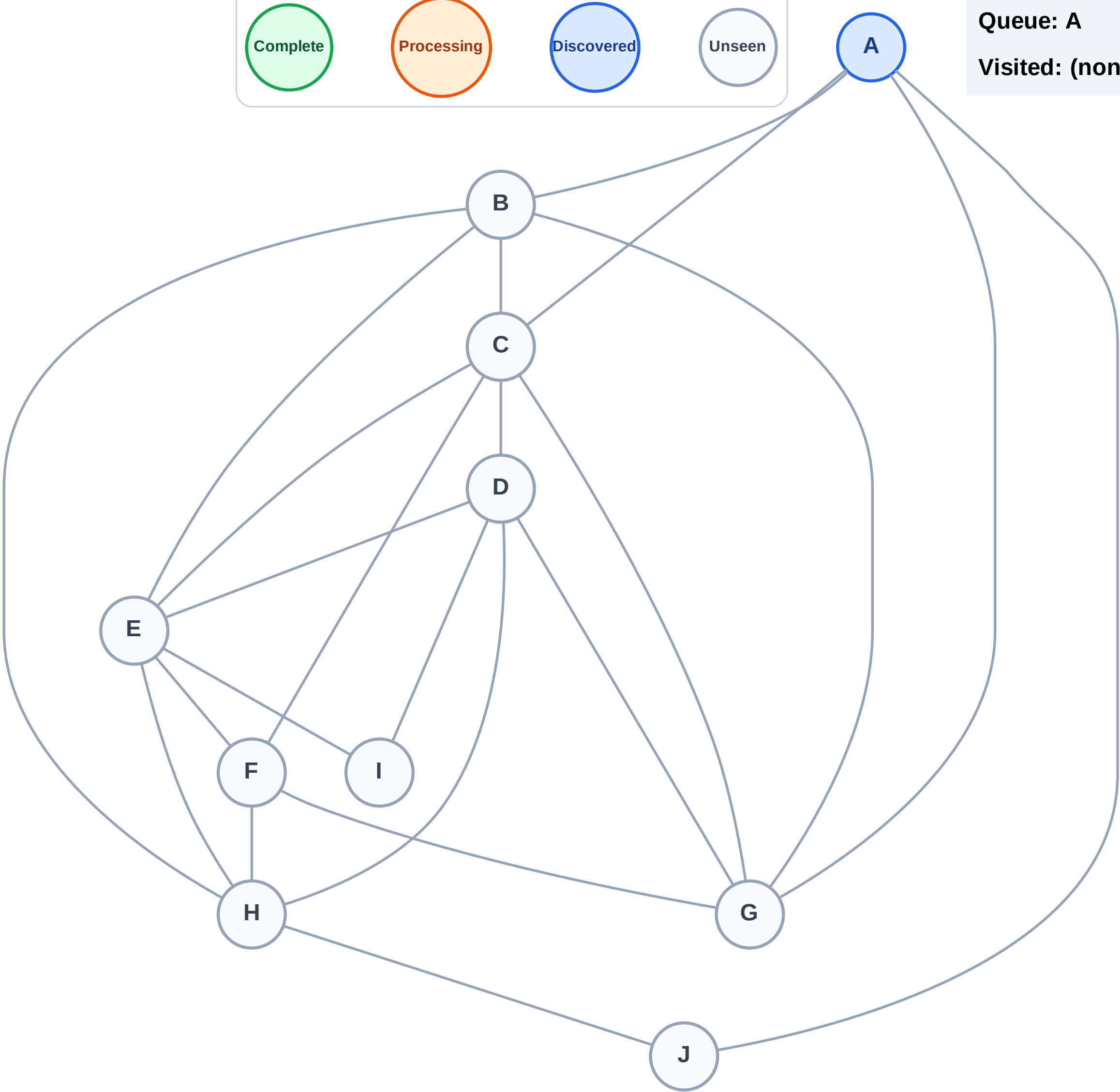
Discovered



Unseen

Queue: A

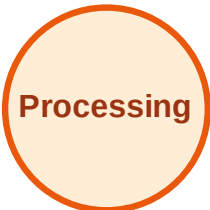
Visited: (none)



BFS Traversal

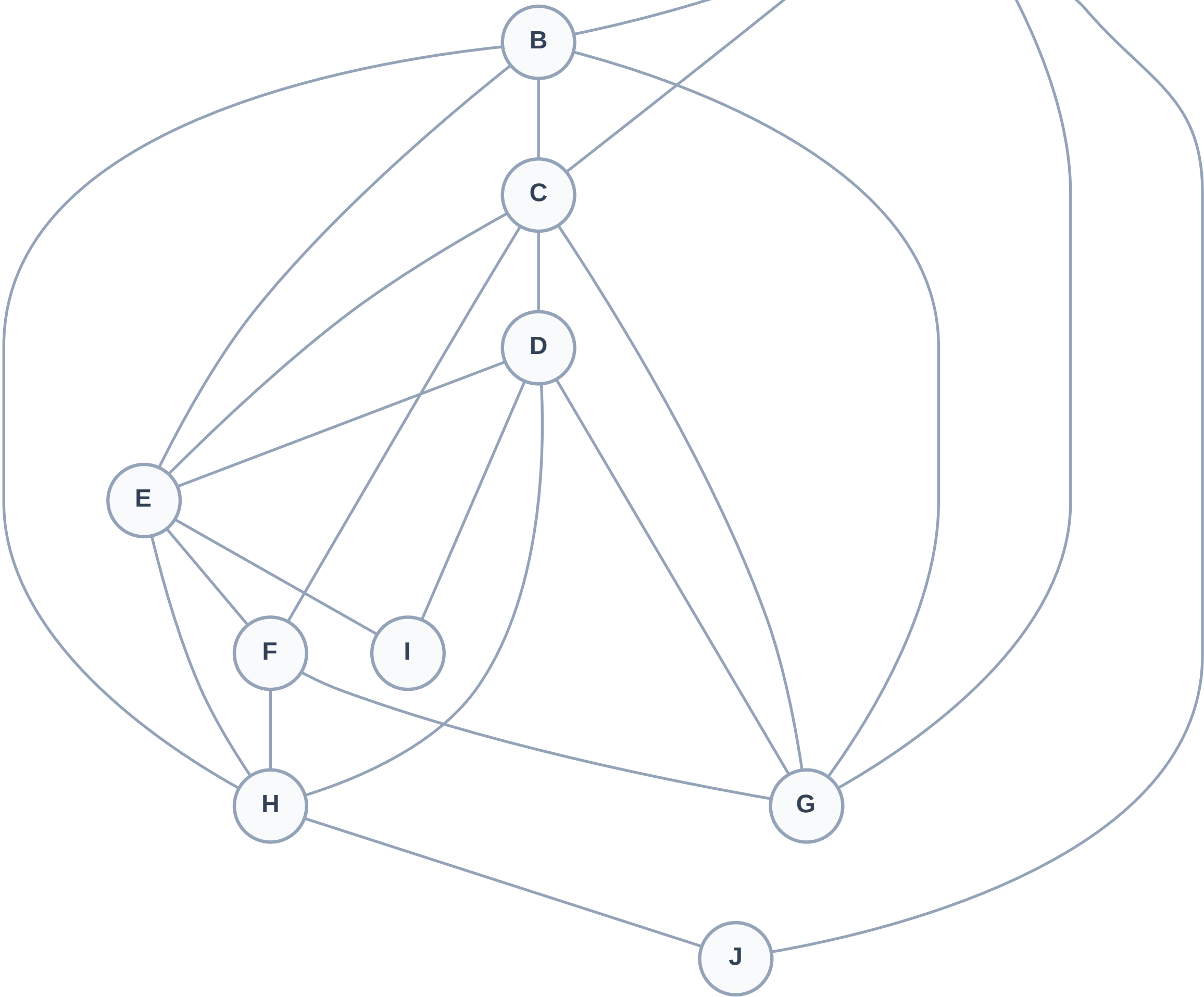
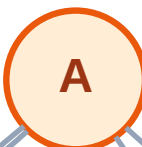
Step 2: Process node A

Legend



Queue: (empty)

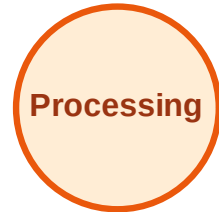
Visited: (none)



BFS Traversal

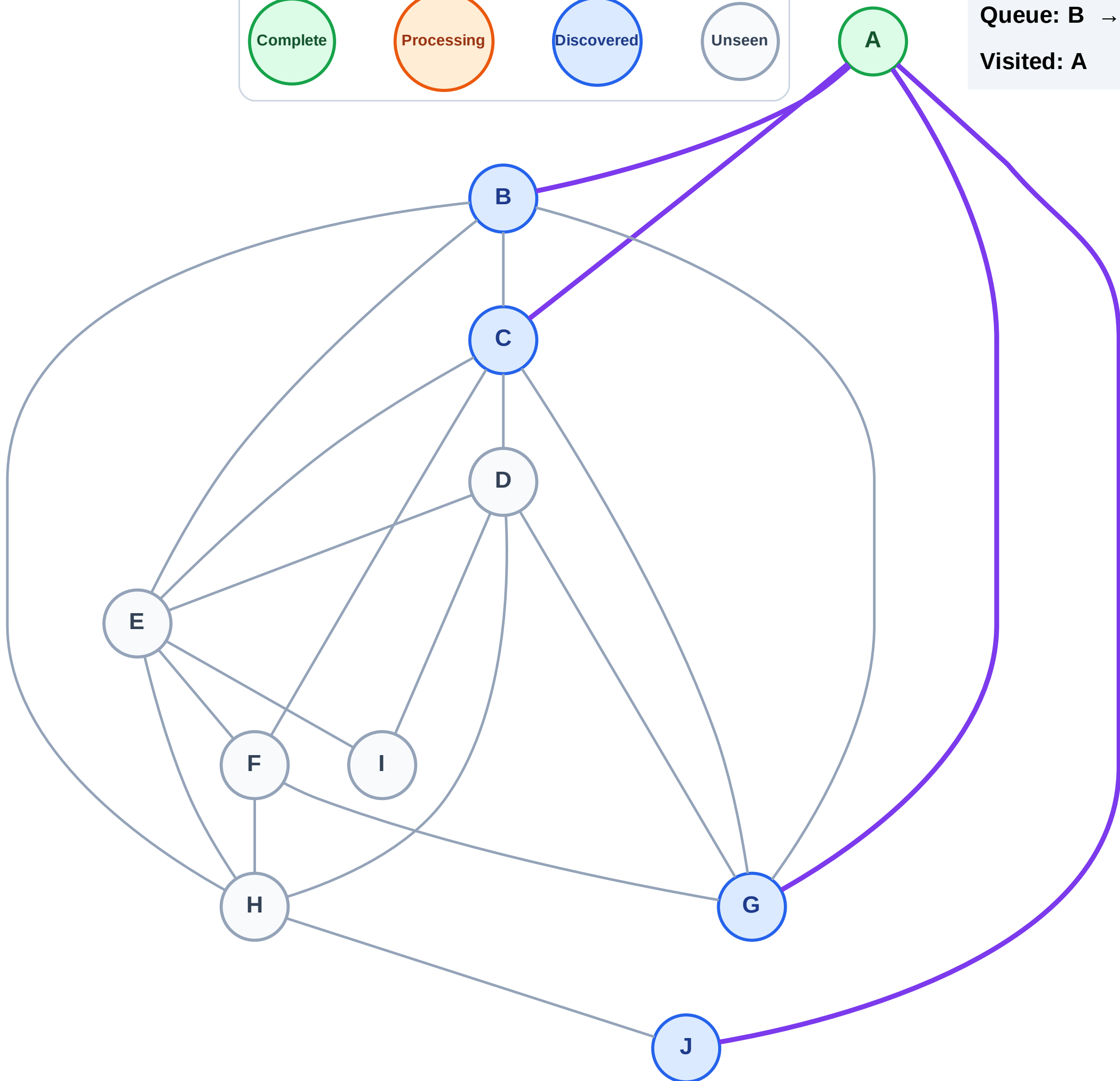
Step 3: Discover B, C, G, J from A

Legend



Queue: B → C → G → J

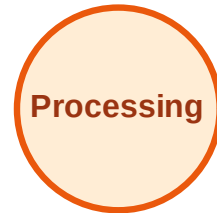
Visited: A



BFS Traversal

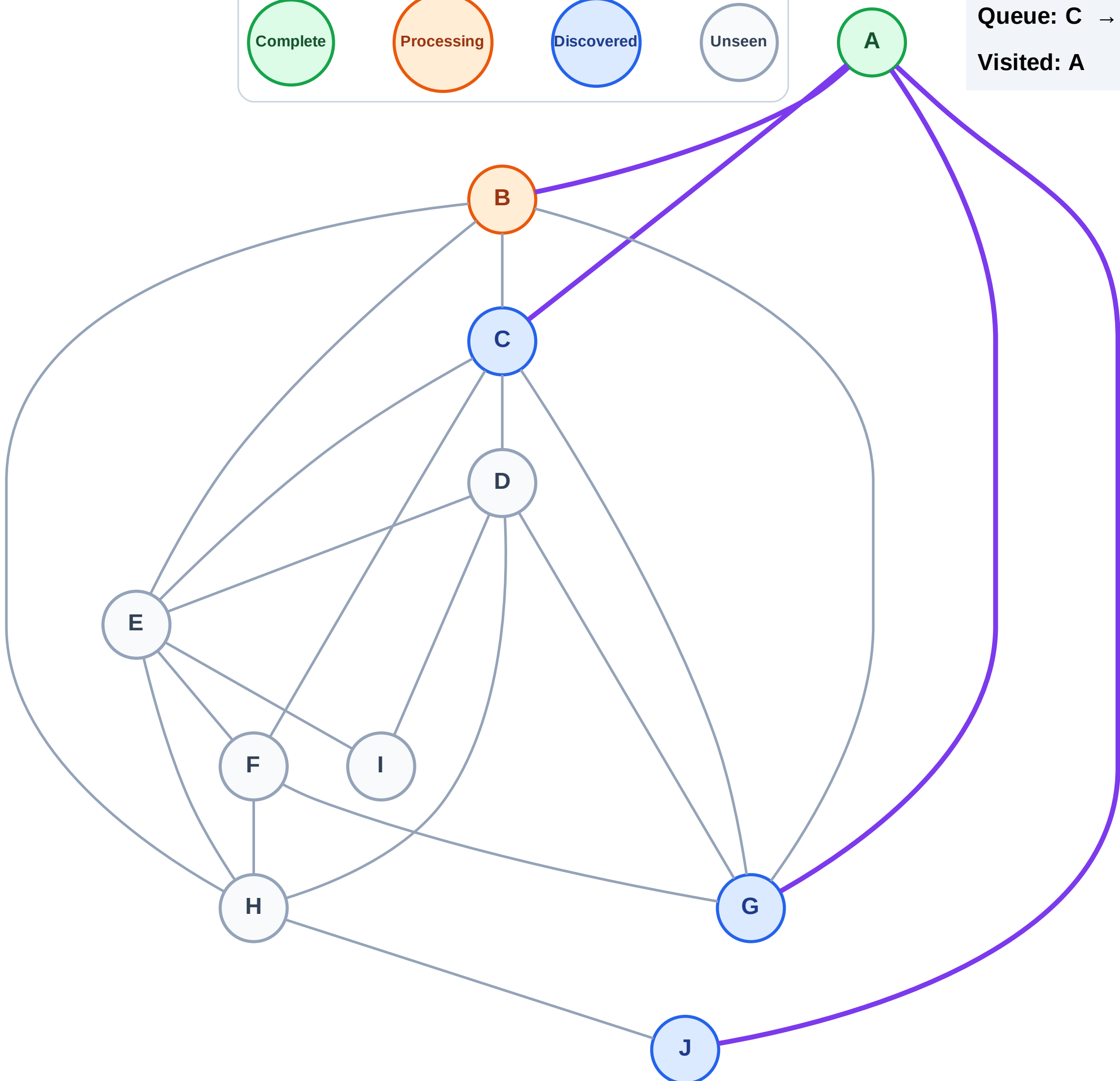
Step 4: Process node B

Legend



Queue: C → G → J

Visited: A



BFS Traversal

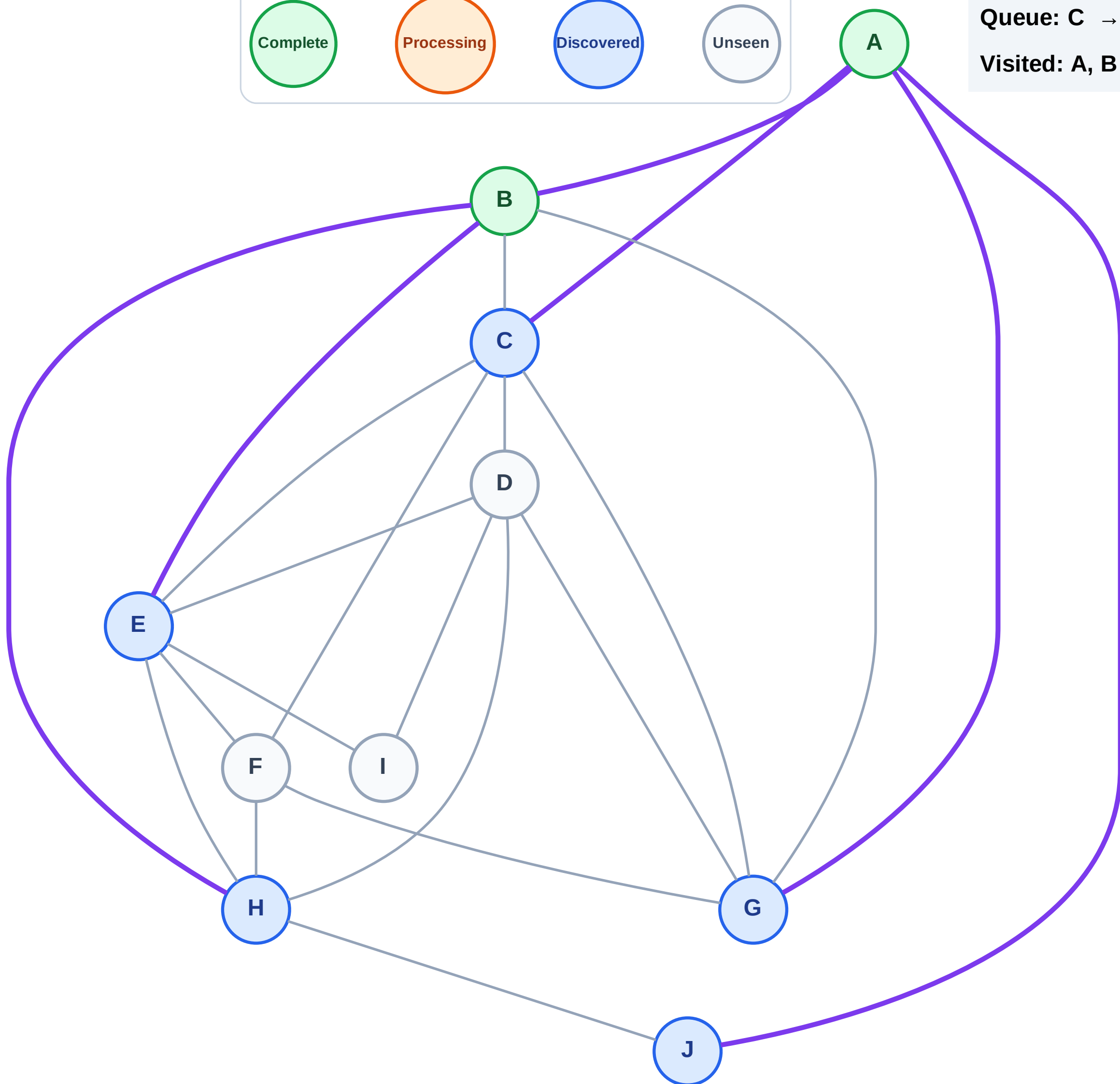
Step 5: Discover E, H from B

Legend



Queue: C → G → J → E → H

Visited: A, B



BFS Traversal

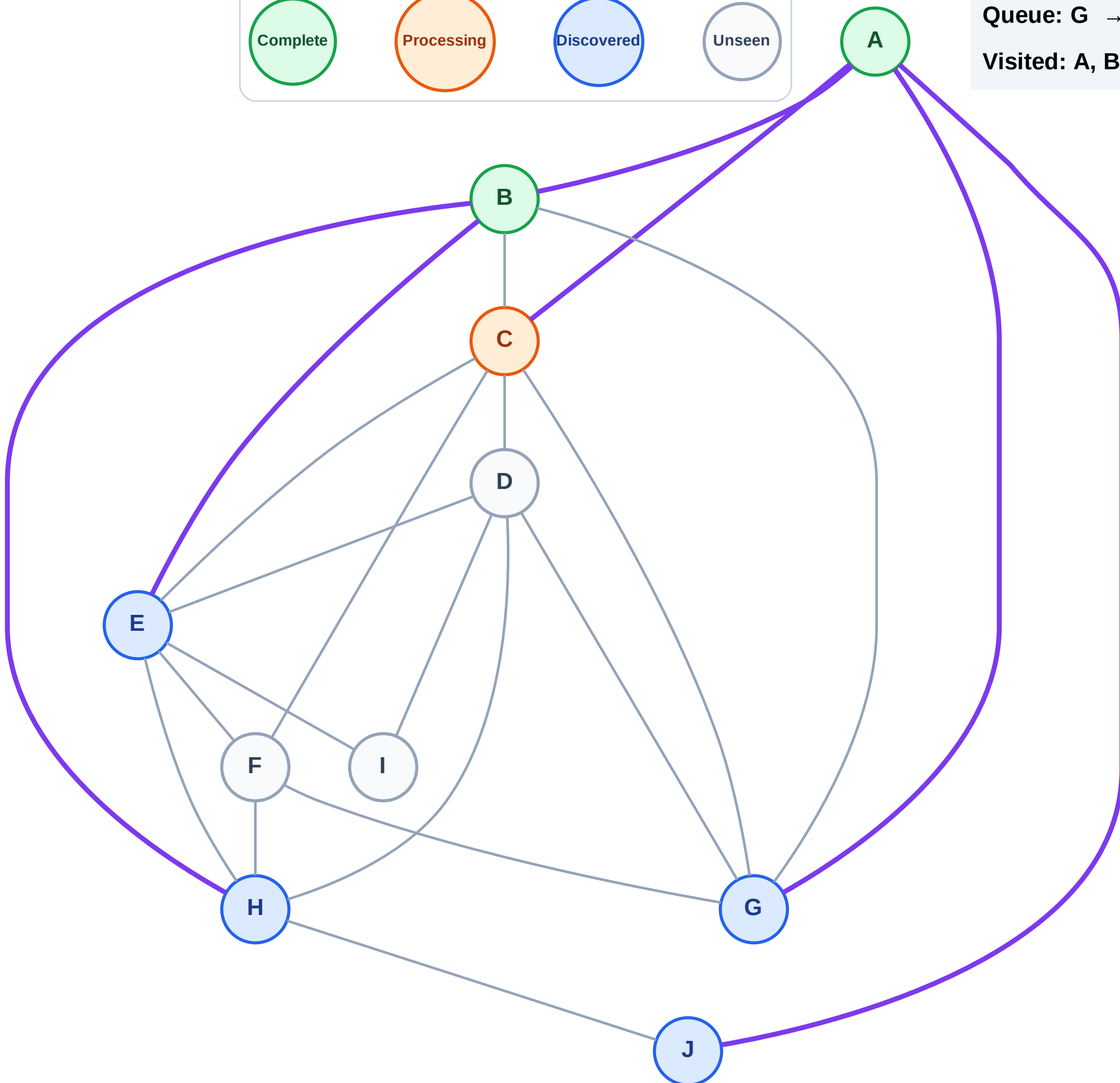
Step 6: Process node C

Legend



Queue: G → J → E → H

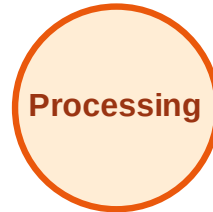
Visited: A, B



BFS Traversal

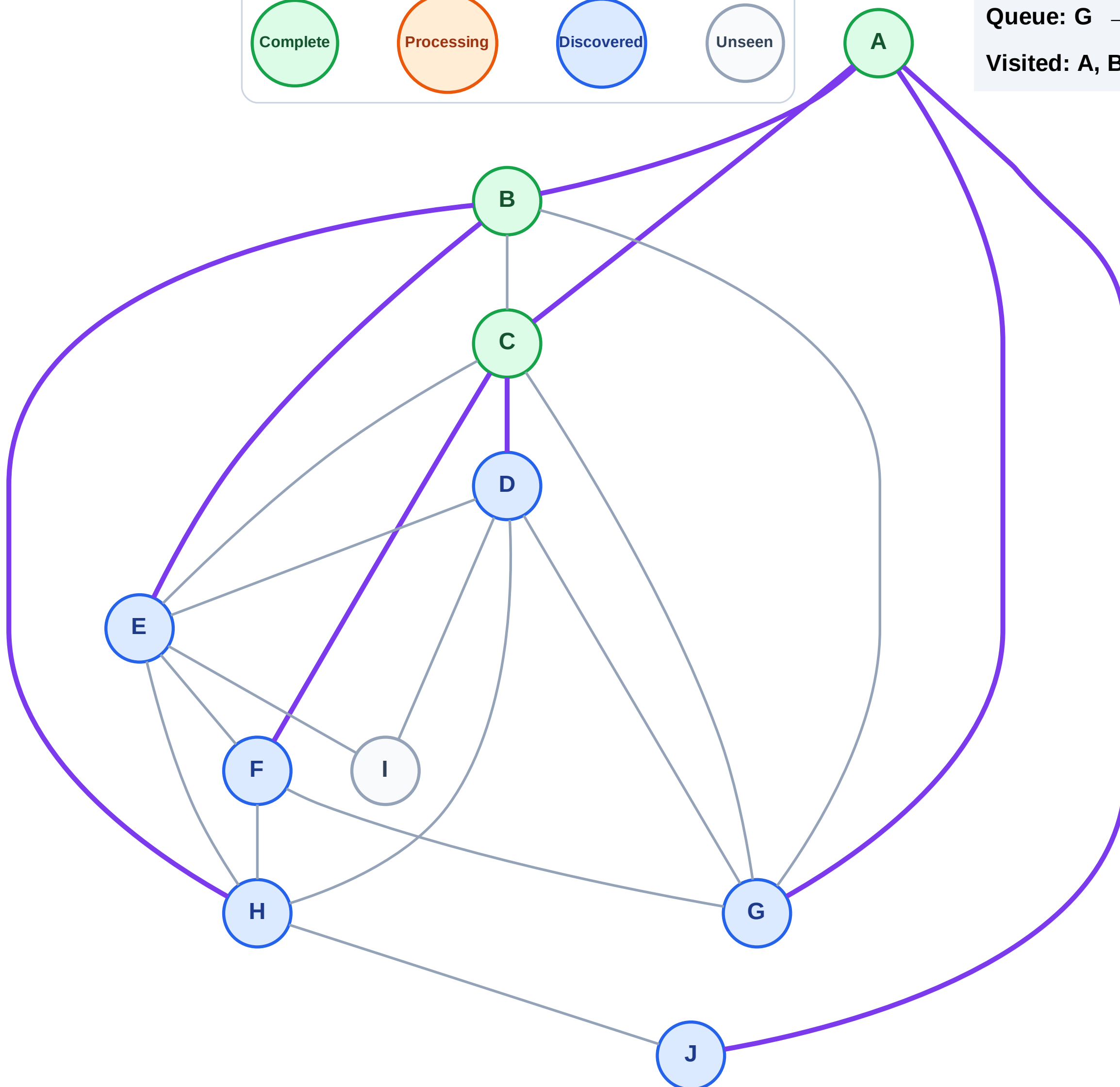
Step 7: Discover D, F from C

Legend



Queue: G → J → E → H → D → F

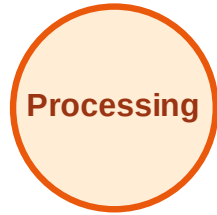
Visited: A, B, C



BFS Traversal

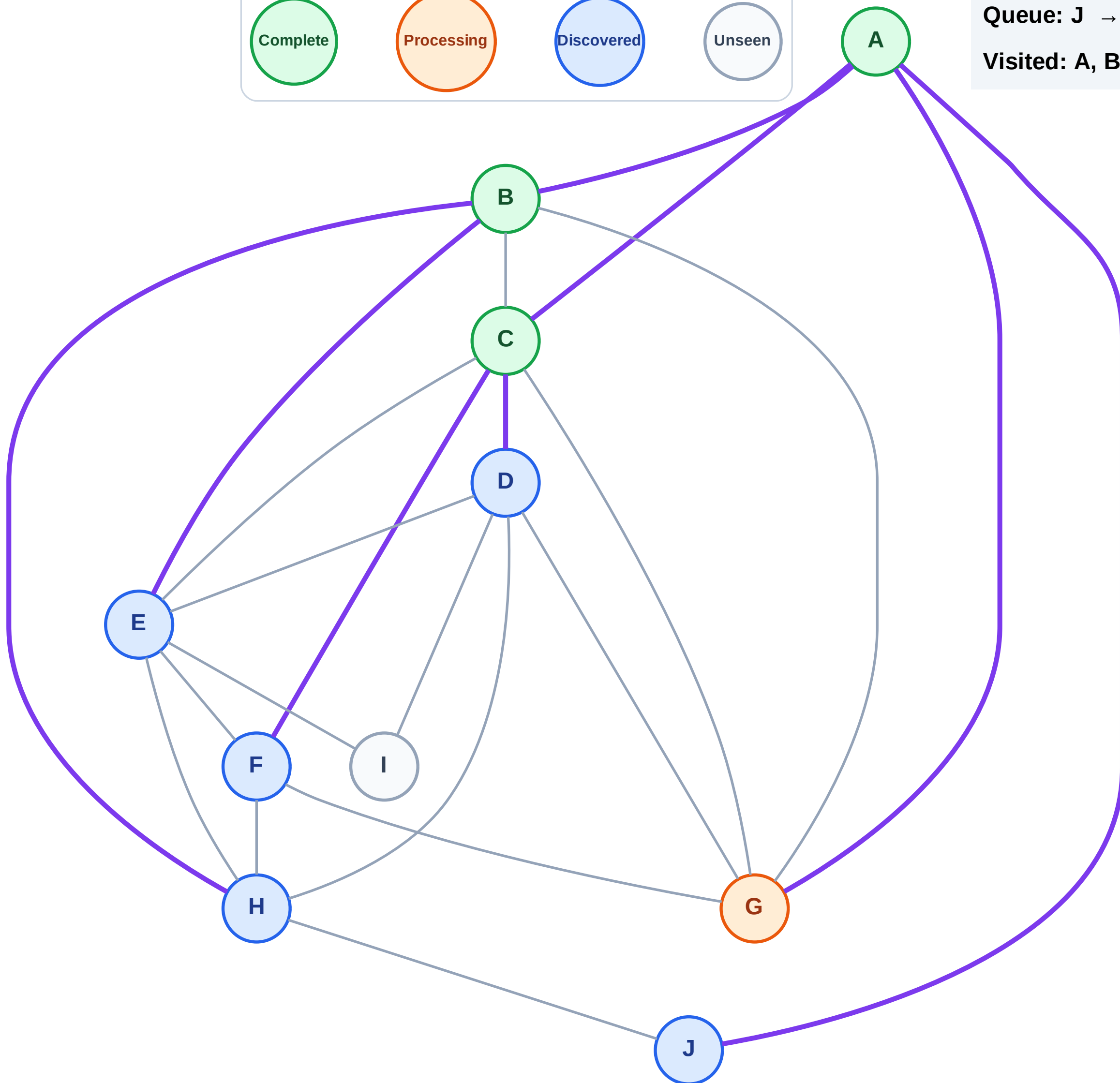
Step 8: Process node G

Legend



Queue: J → E → H → D → F

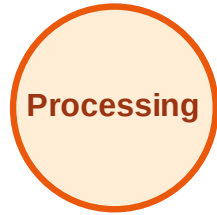
Visited: A, B, C



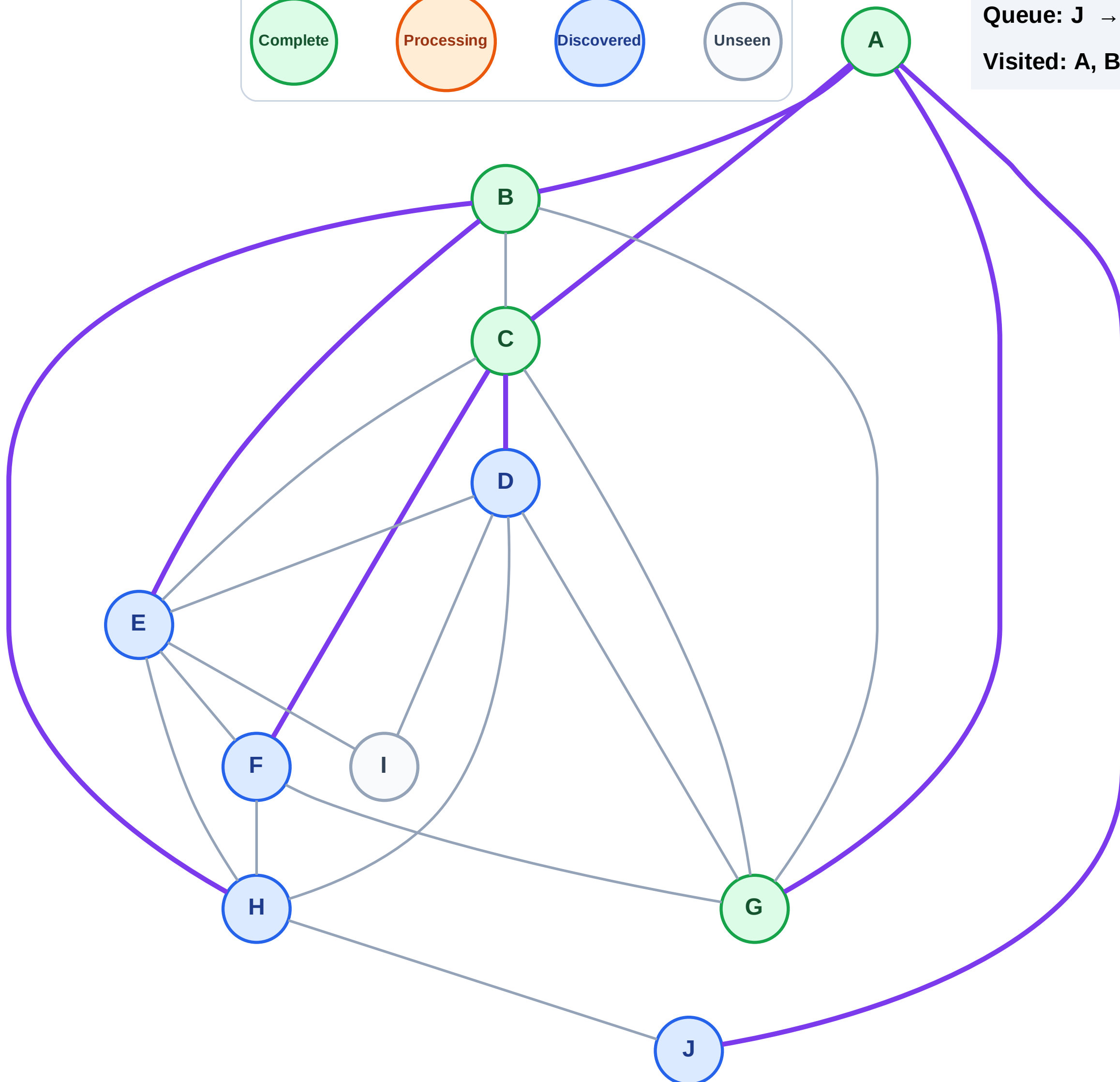
BFS Traversal

Step 9: No new neighbors from G

Legend



Queue: J → E → H → D → F
Visited: A, B, C, G



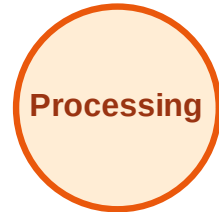
BFS Traversal

Step 10: Process node J

Legend



Complete



Processing



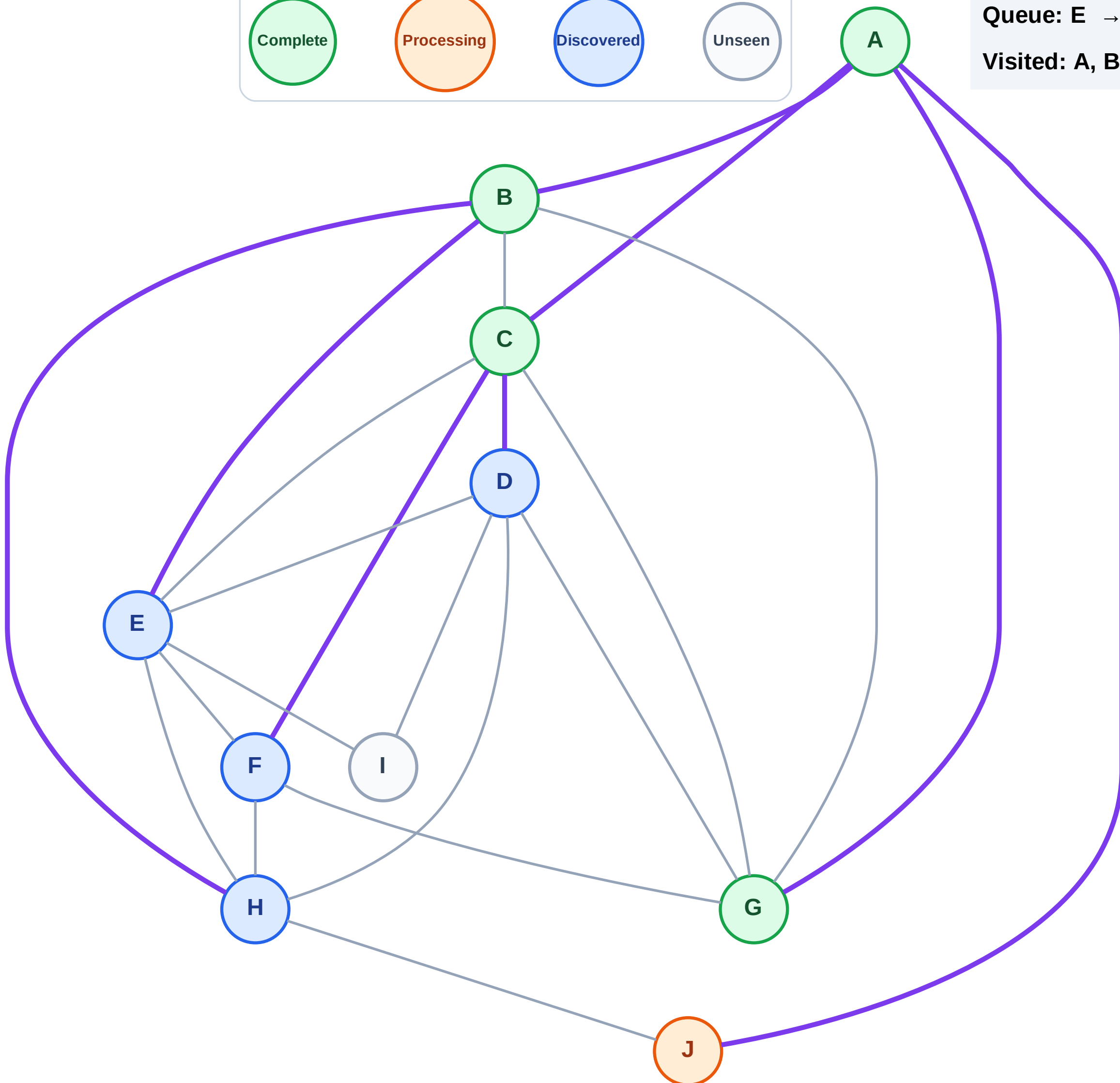
Discovered



Unseen

Queue: E → H → D → F

Visited: A, B, C, G



BFS Traversal

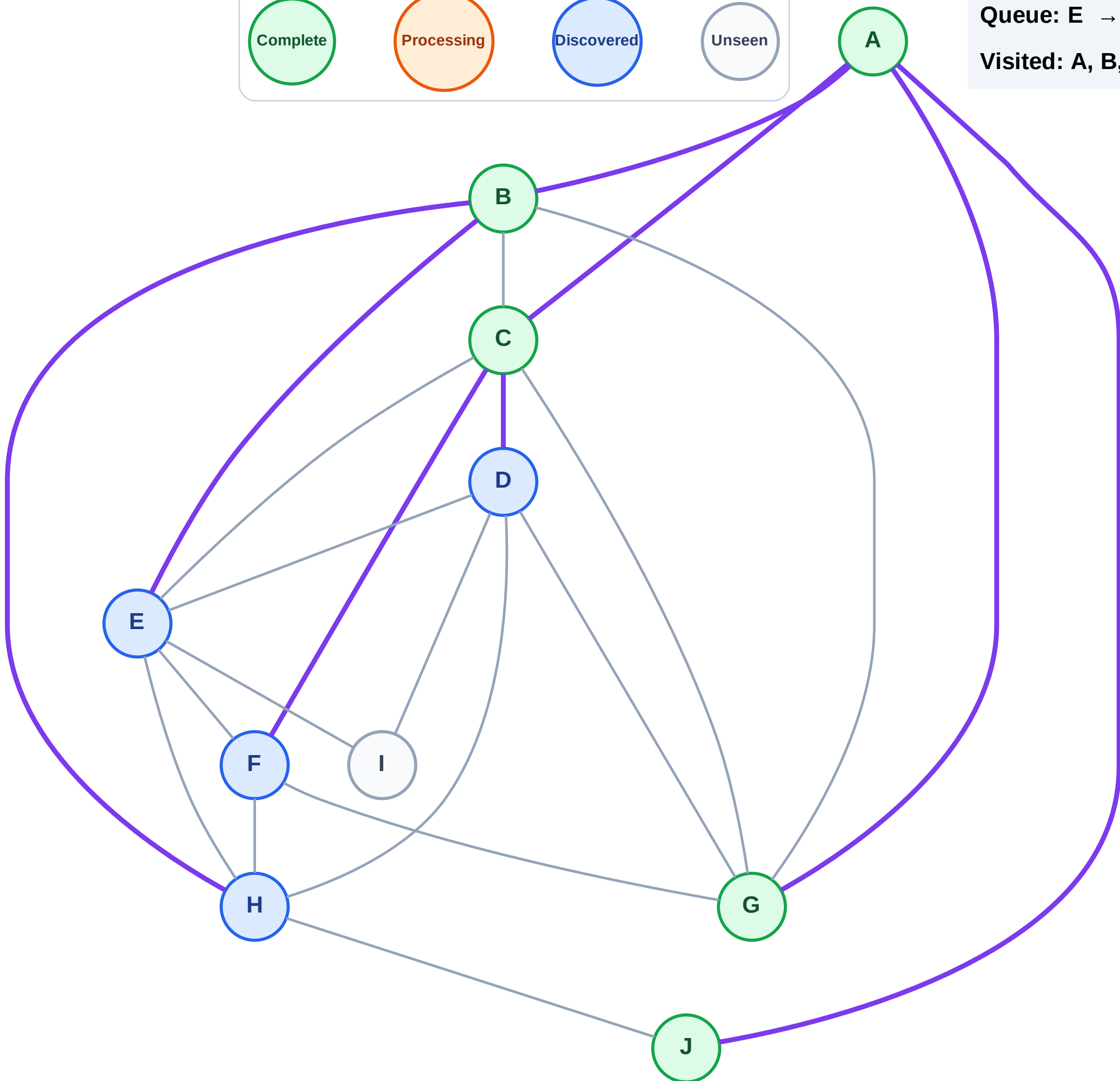
Step 11: No new neighbors from J

Legend



Queue: E → H → D → F

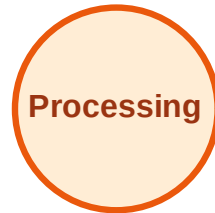
Visited: A, B, C, G, J



BFS Traversal

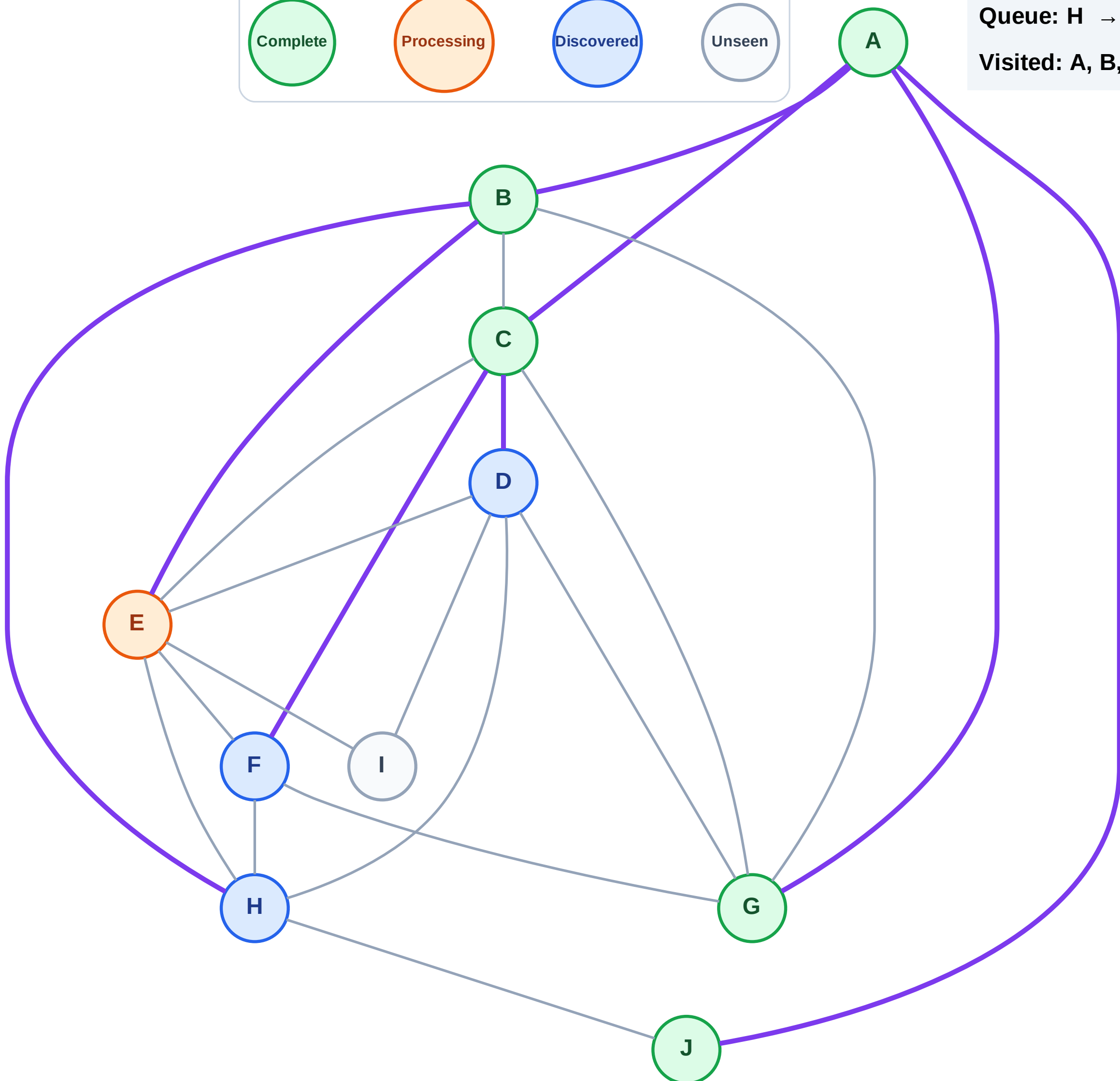
Step 12: Process node E

Legend



Queue: H → D → F

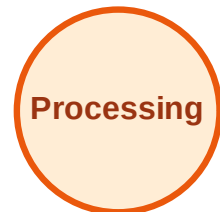
Visited: A, B, C, G, J



BFS Traversal

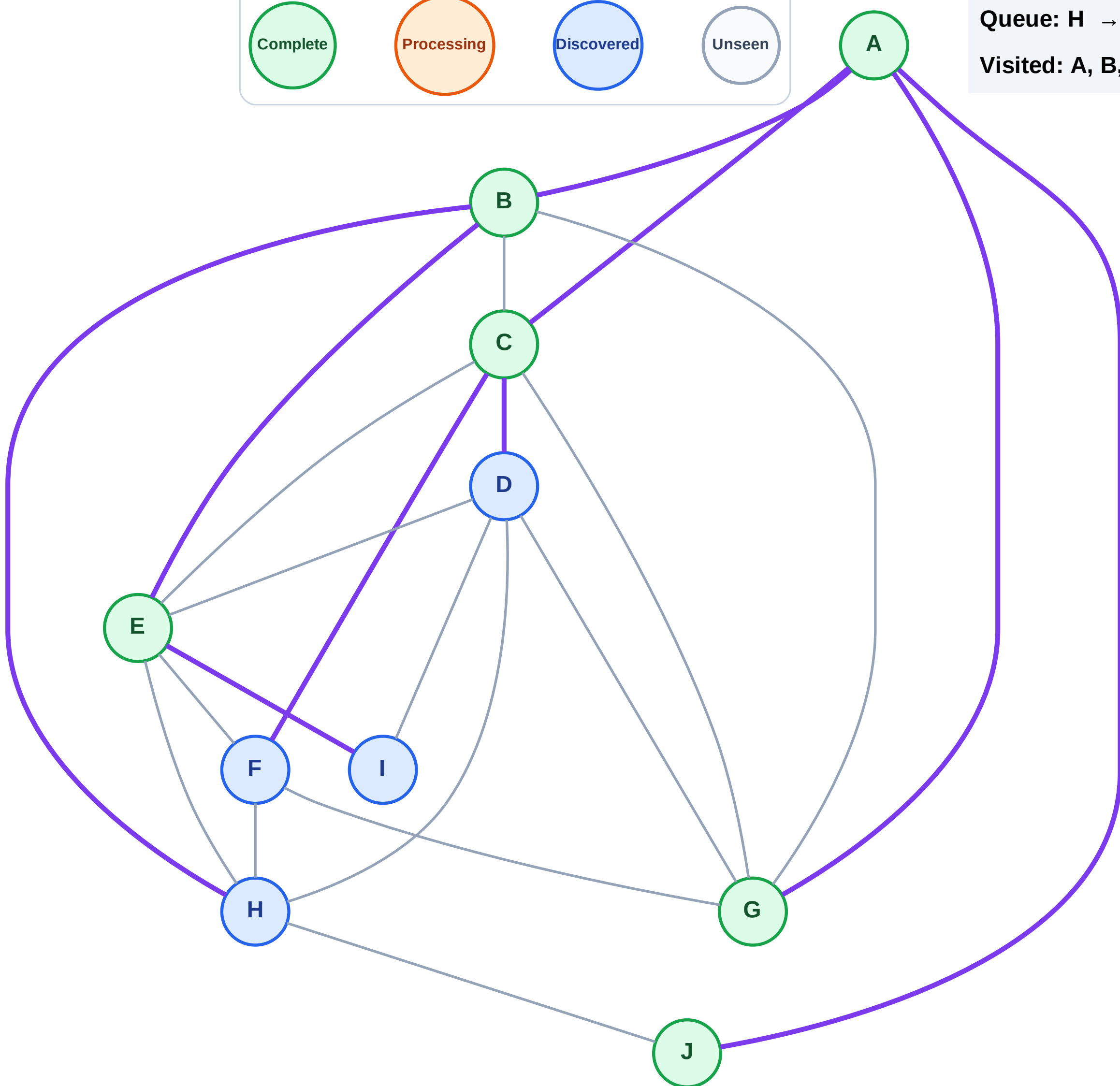
Step 13: Discover I from E

Legend



Queue: H → D → F → I

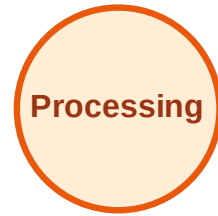
Visited: A, B, C, E, G, J



BFS Traversal

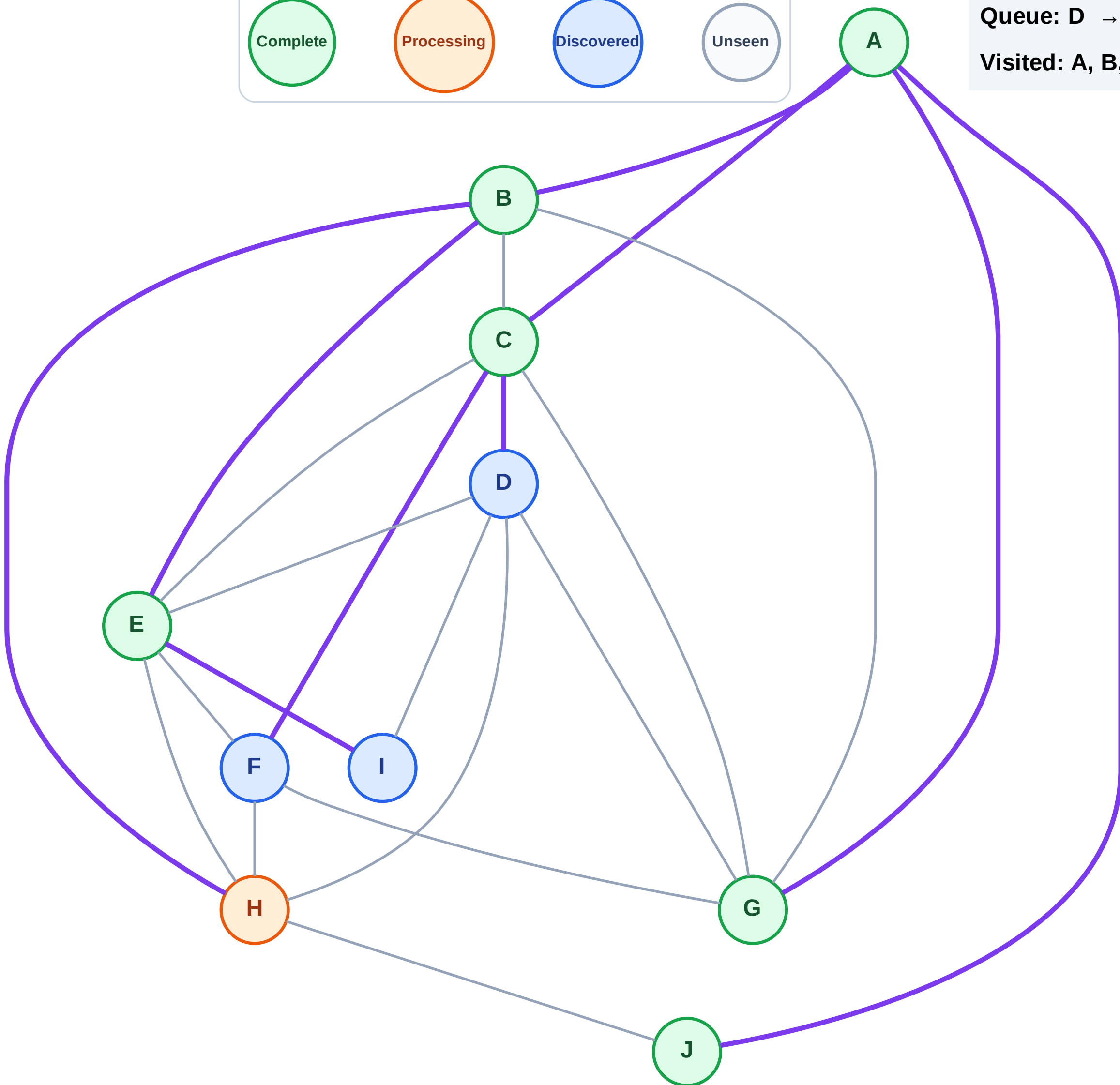
Step 14: Process node H

Legend



Queue: D → F → I

Visited: A, B, C, E, G, J



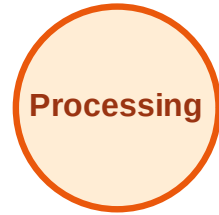
BFS Traversal

Step 15: No new neighbors from H

Legend



Complete



Processing



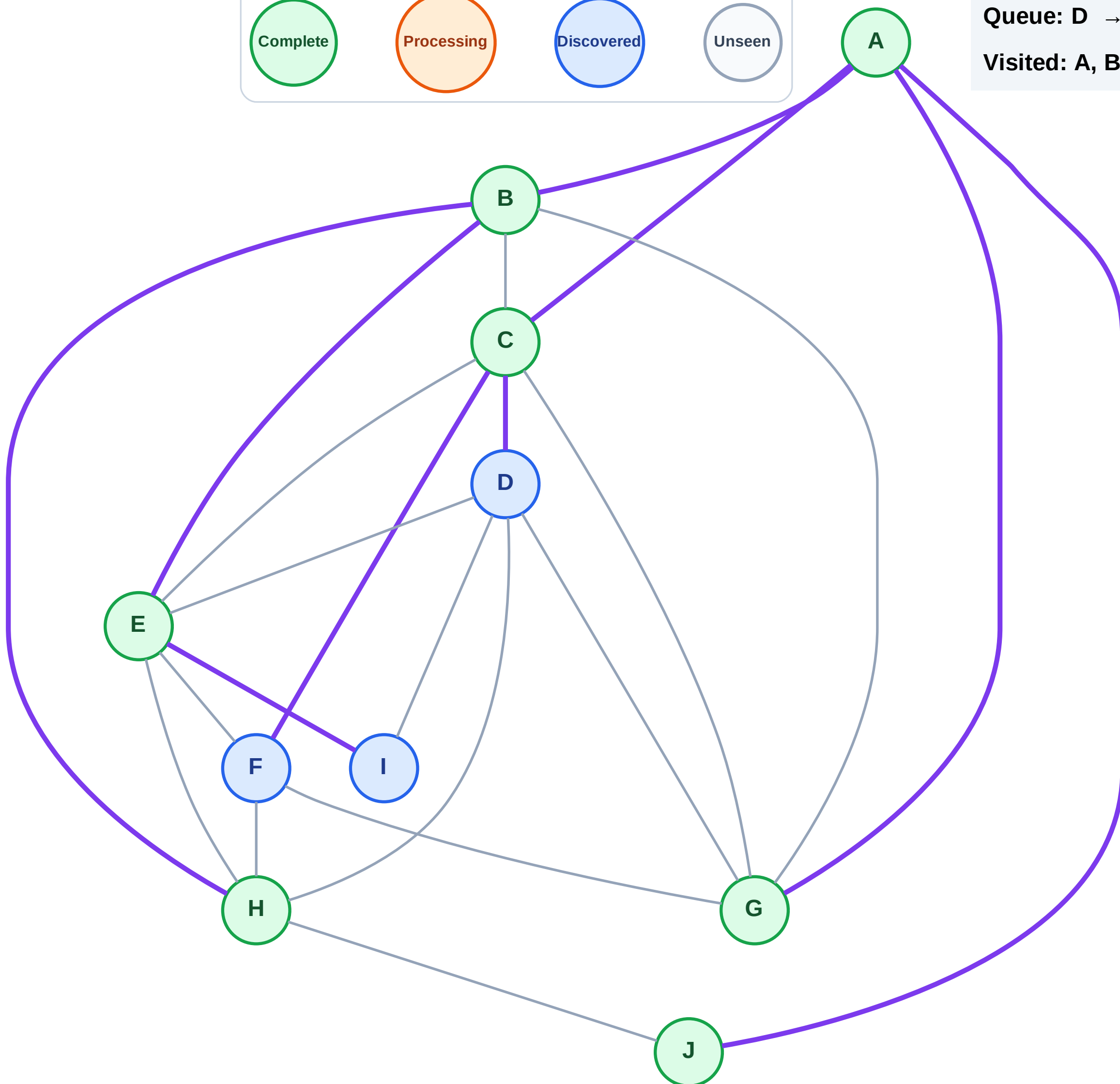
Discovered



Unseen

Queue: D → F → I

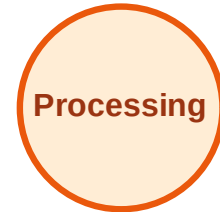
Visited: A, B, C, E, G, H, J



BFS Traversal

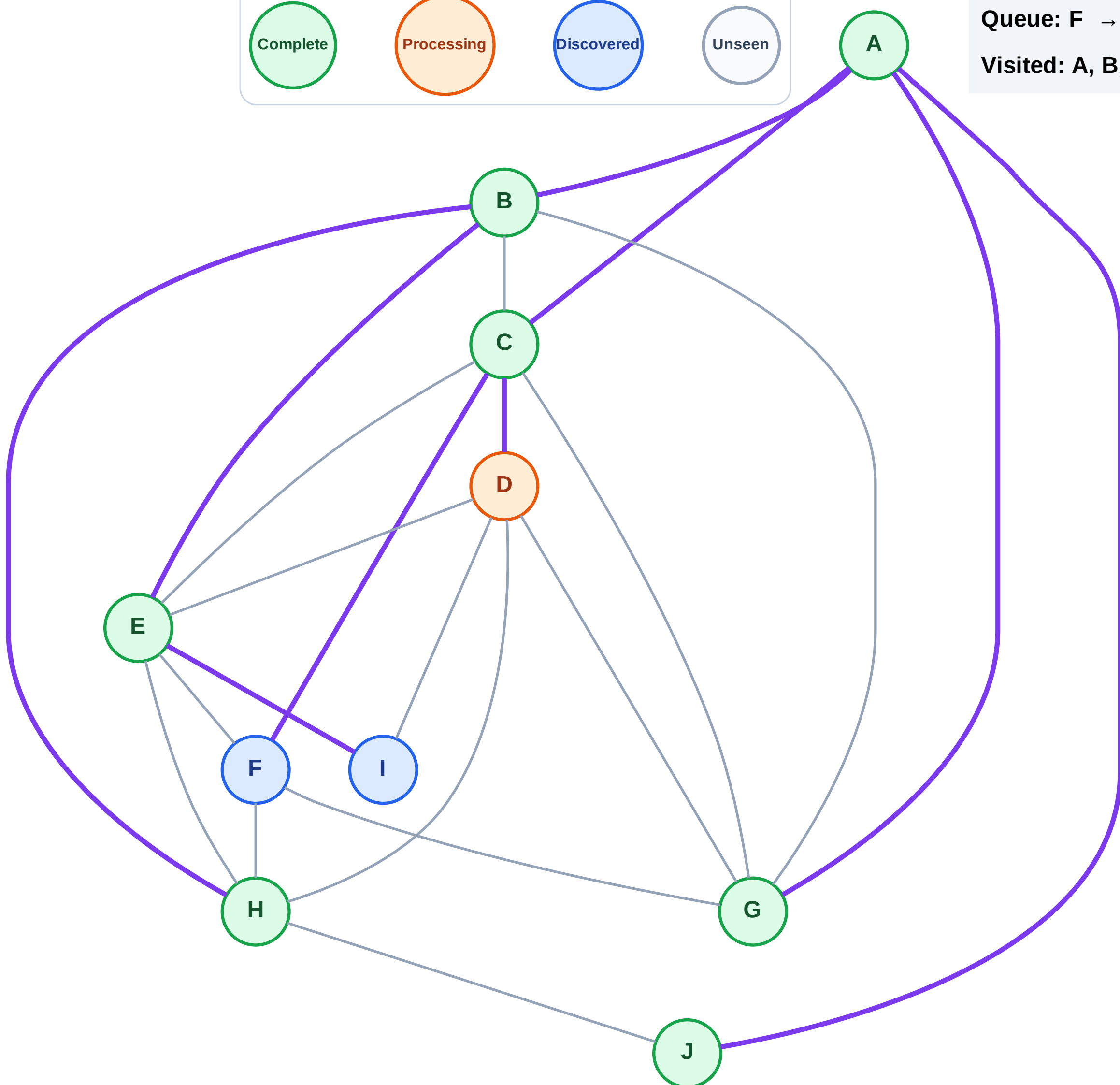
Step 16: Process node D

Legend



Queue: F → I

Visited: A, B, C, E, G, H, J



BFS Traversal

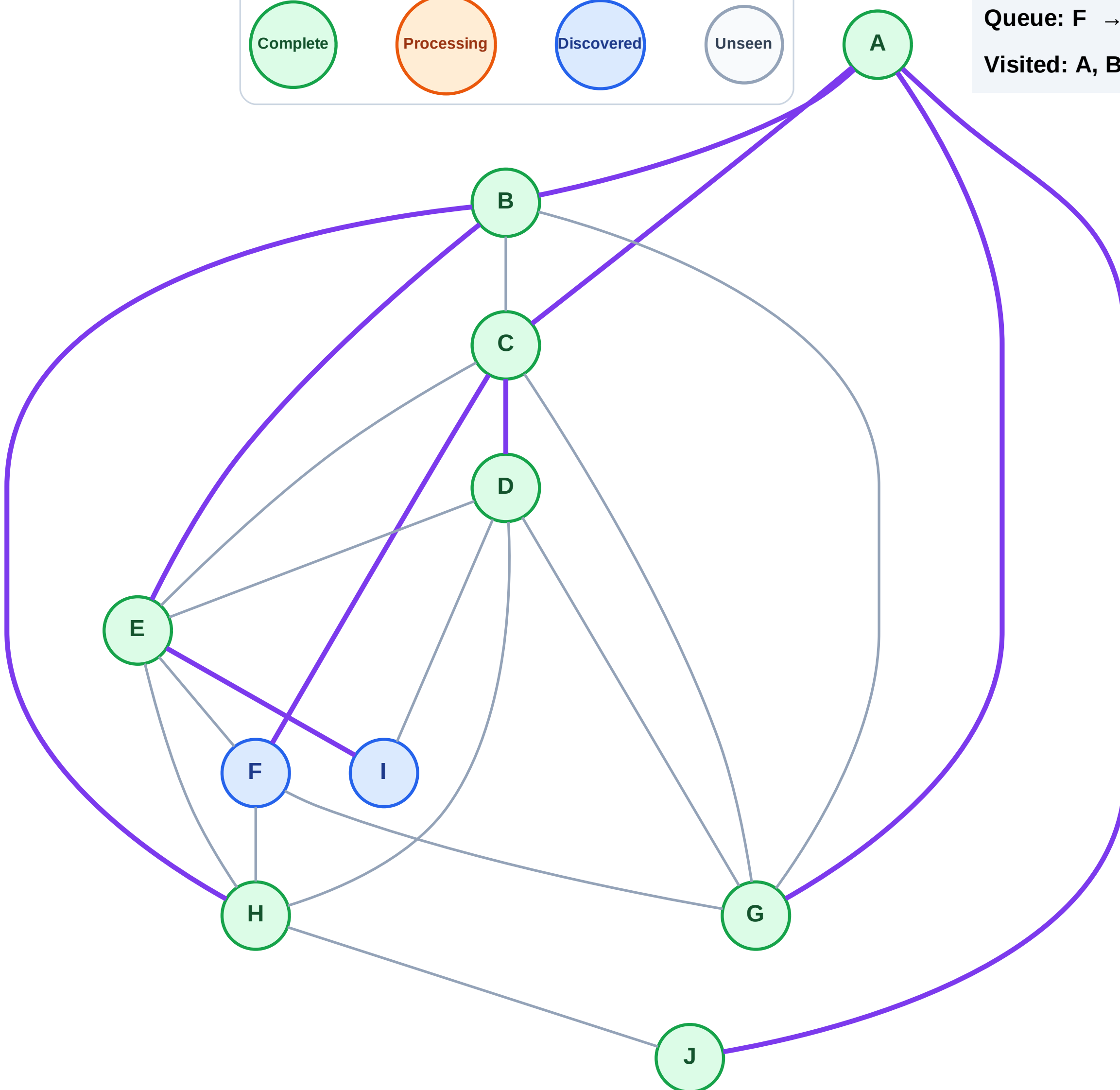
Step 17: No new neighbors from D

Legend



Queue: F → I

Visited: A, B, C, D, E, G, H, J



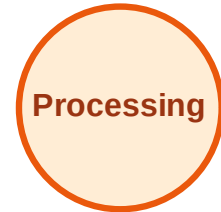
BFS Traversal

Step 18: Process node F

Legend



Complete



Processing



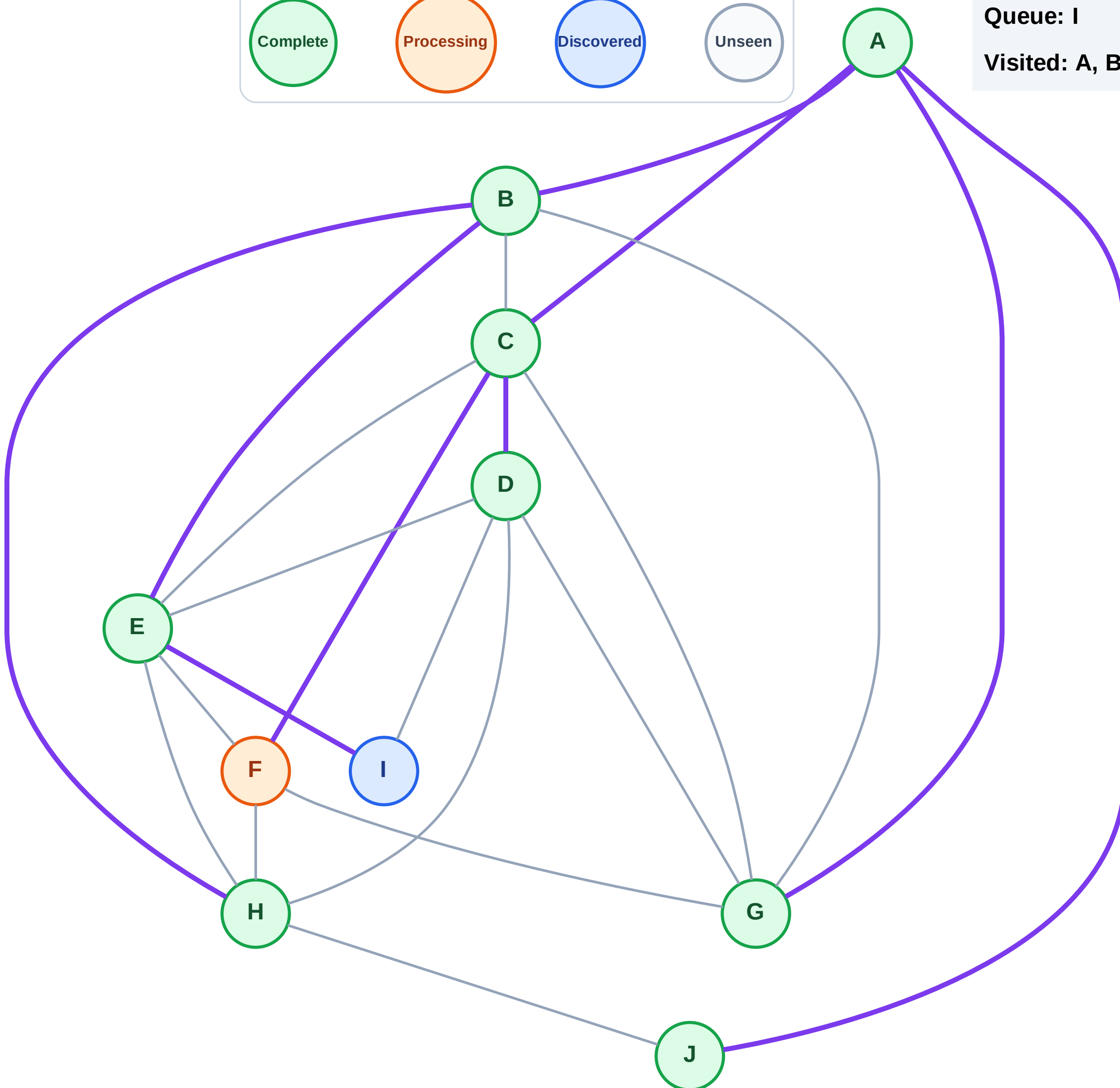
Discovered



Unseen

Queue: I

Visited: A, B, C, D, E, G, H, J



BFS Traversal

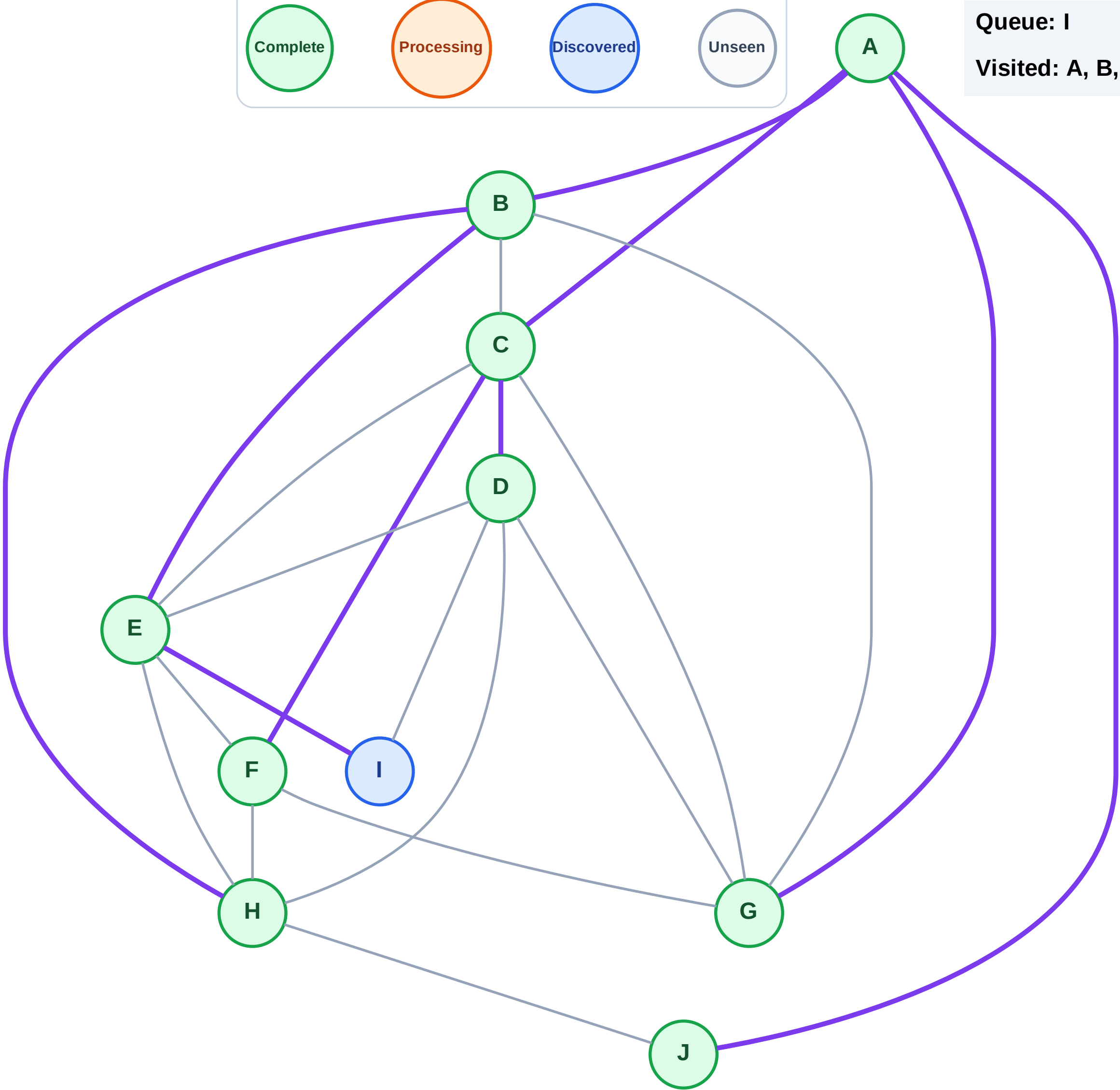
Step 19: No new neighbors from F

Legend



Queue: I

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

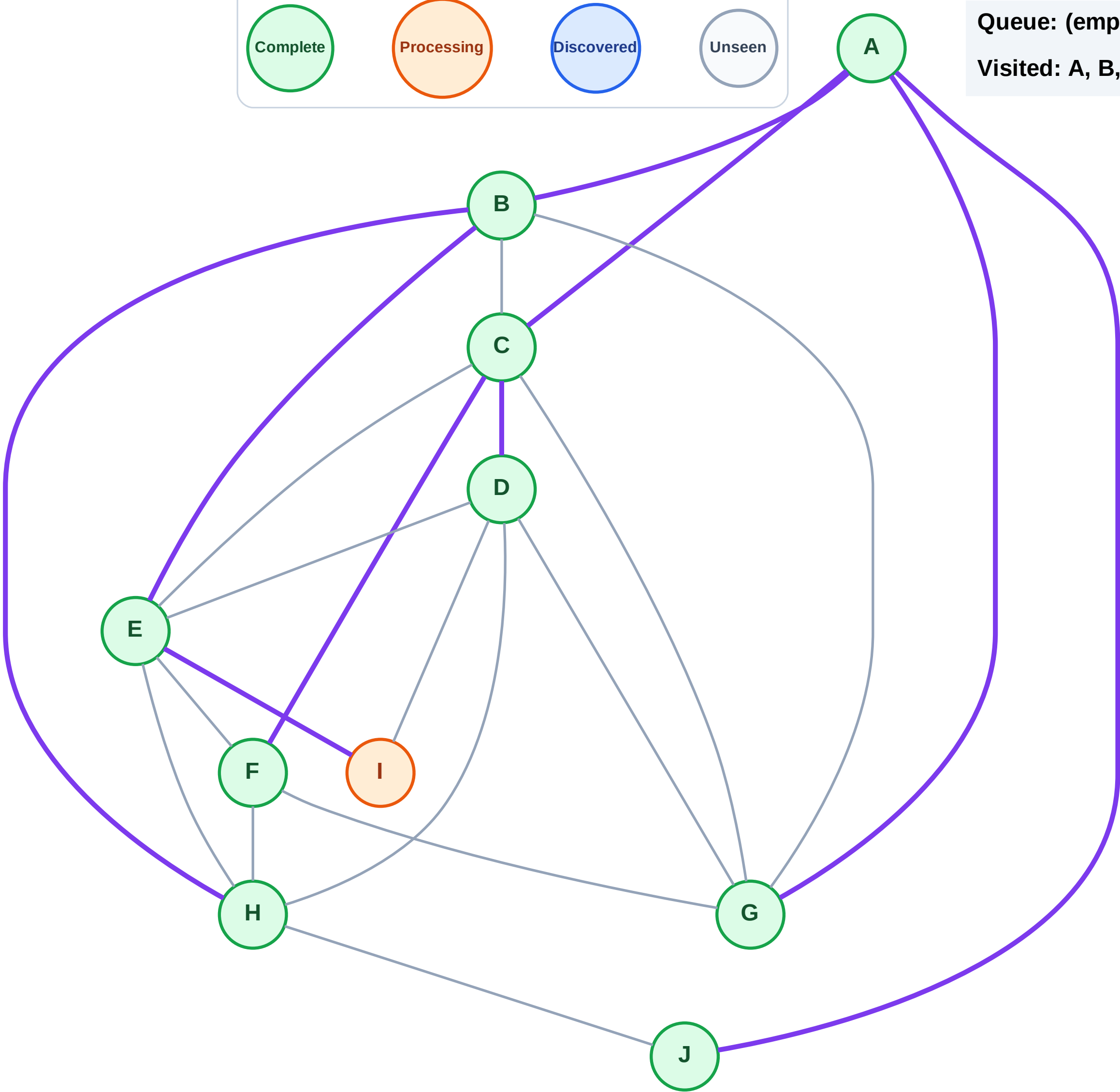
Step 20: Process node I

Legend



Queue: (empty)

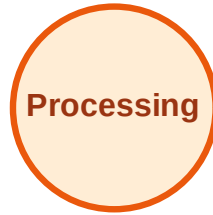
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

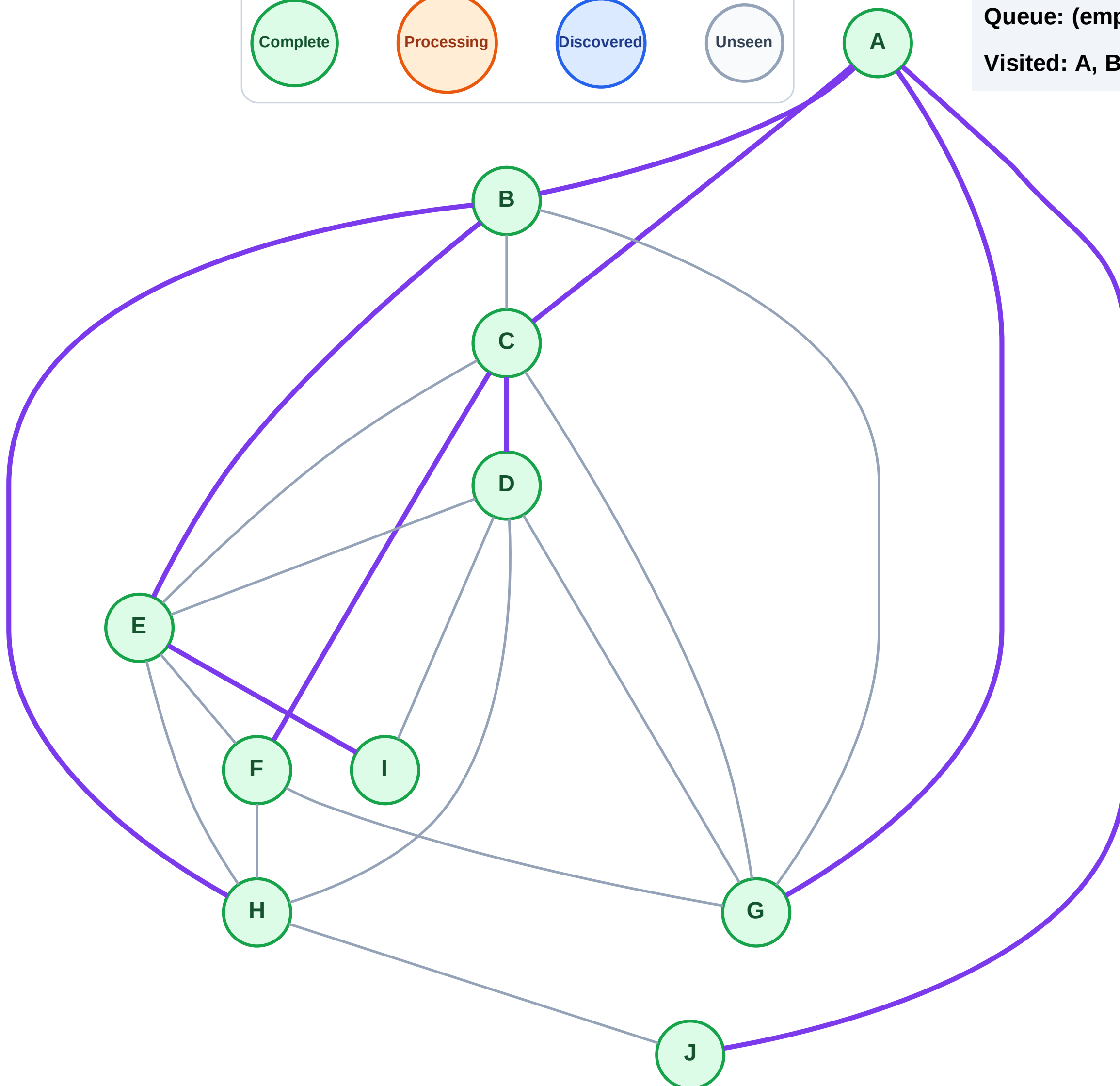
Step 21: No new neighbors from I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



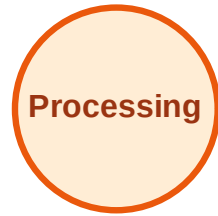
DFS Traversal

Step 1: Start at A

Legend



Complete



Processing



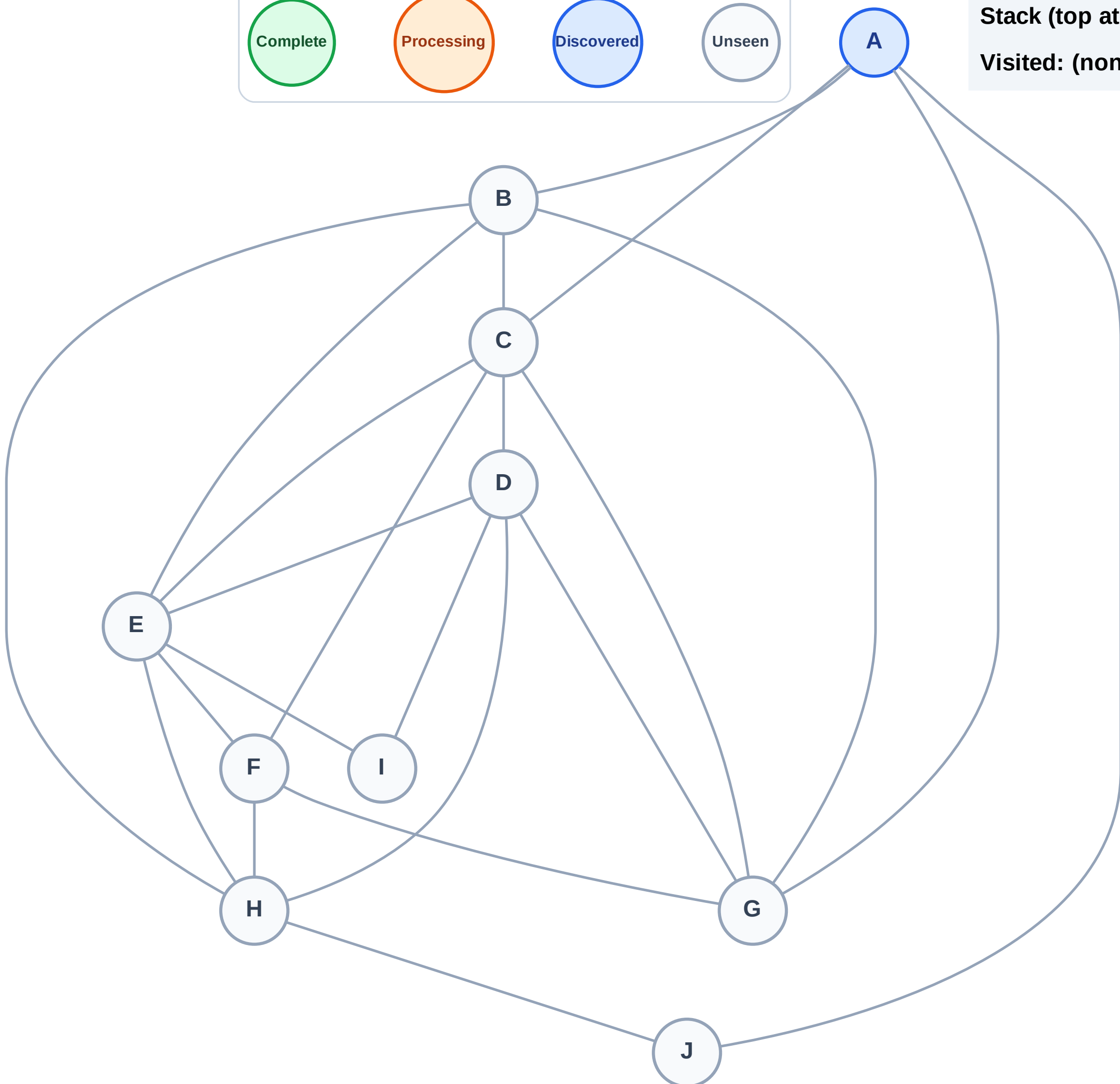
Discovered



Unseen

Stack (top at right): A

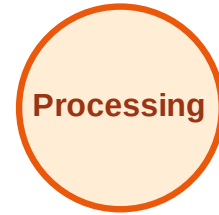
Visited: (none)



DFS Traversal

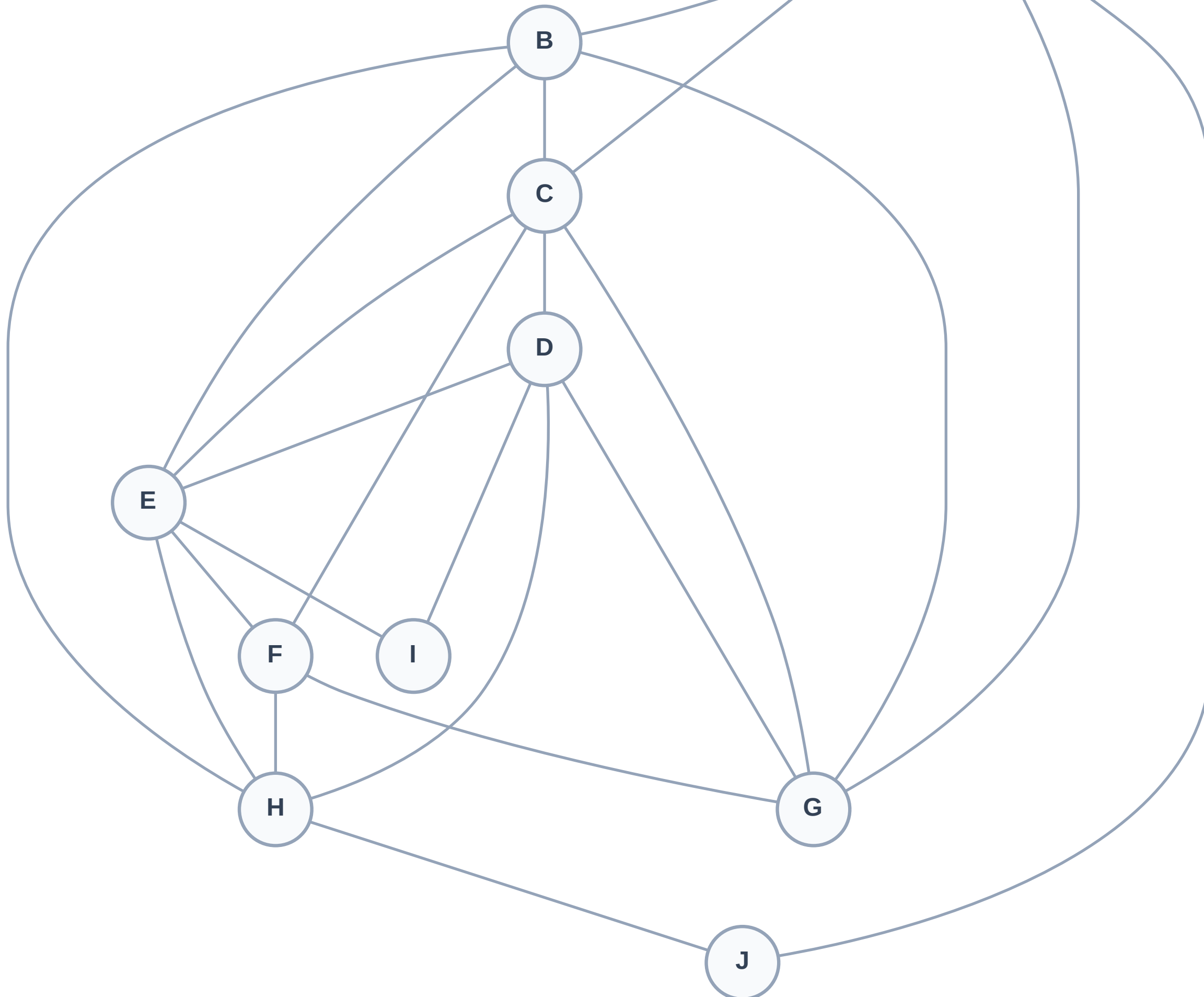
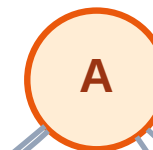
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

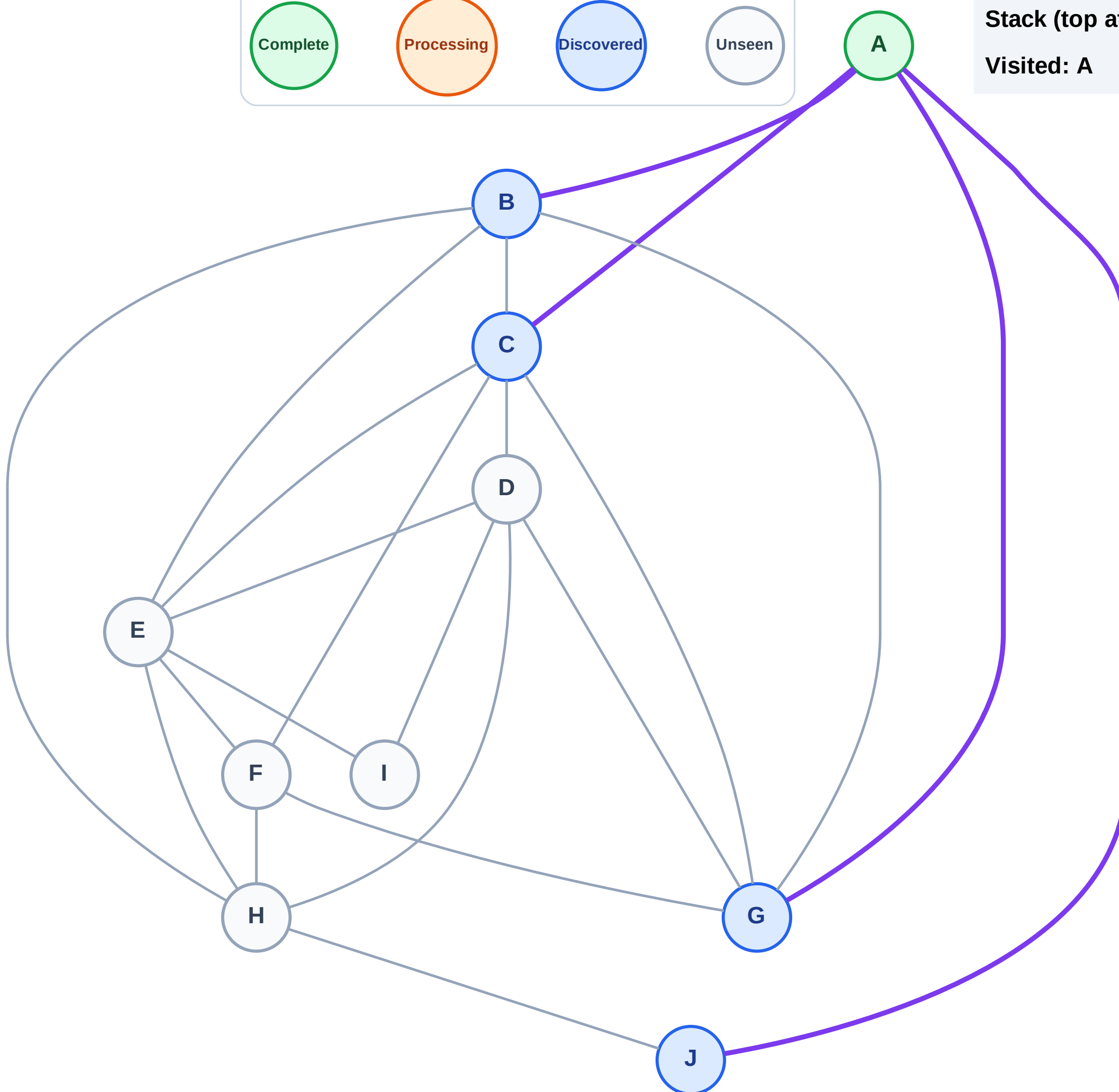
Step 3: Discover J, G, C, B from A

Legend



Stack (top at right): J | G | C | B

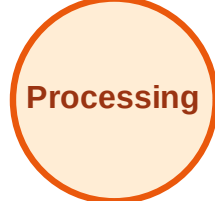
Visited: A



DFS Traversal

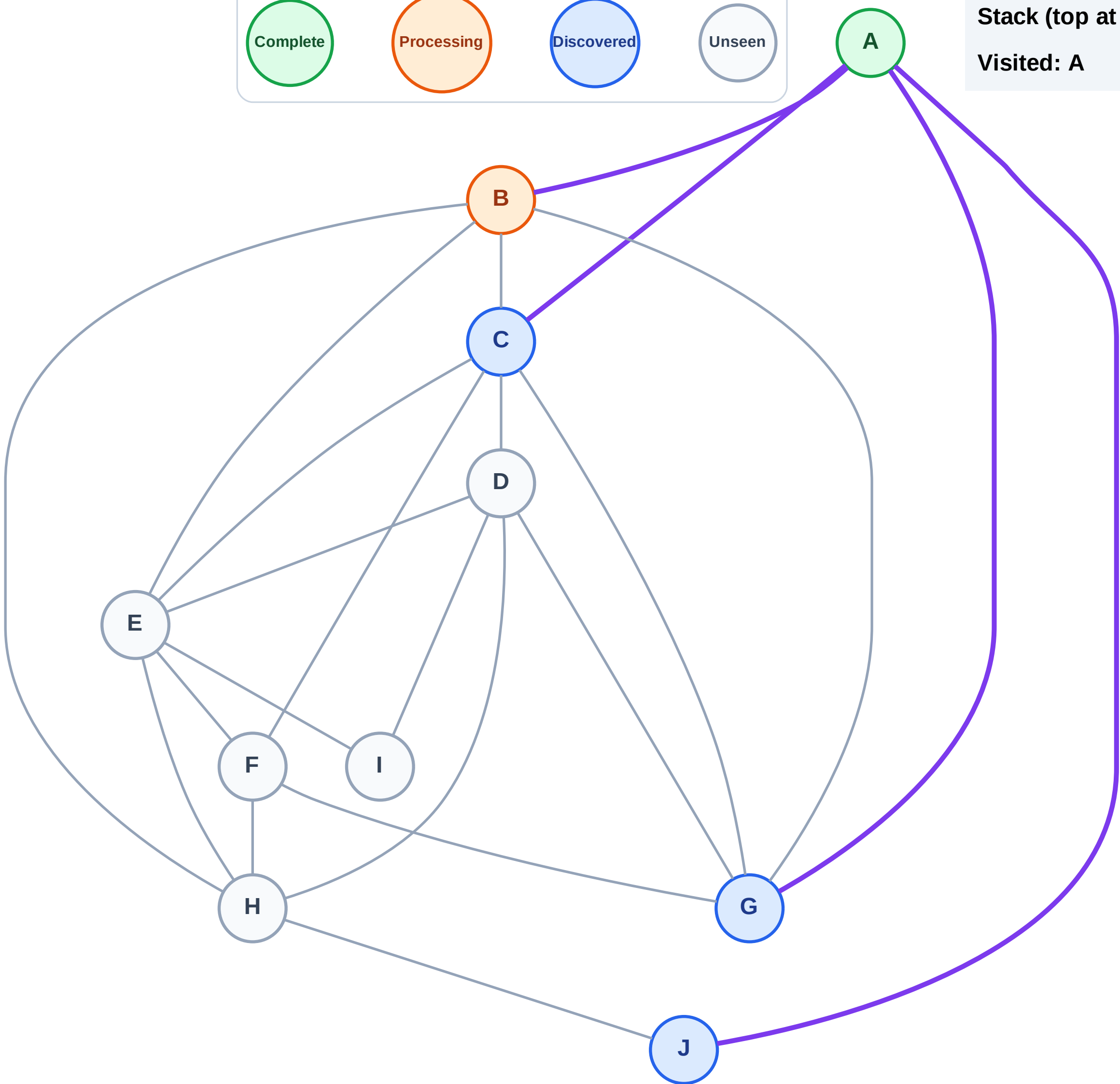
Step 4: Process node B

Legend



Stack (top at right): J | G | C

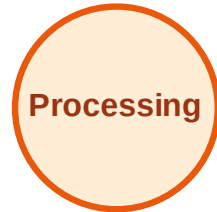
Visited: A



DFS Traversal

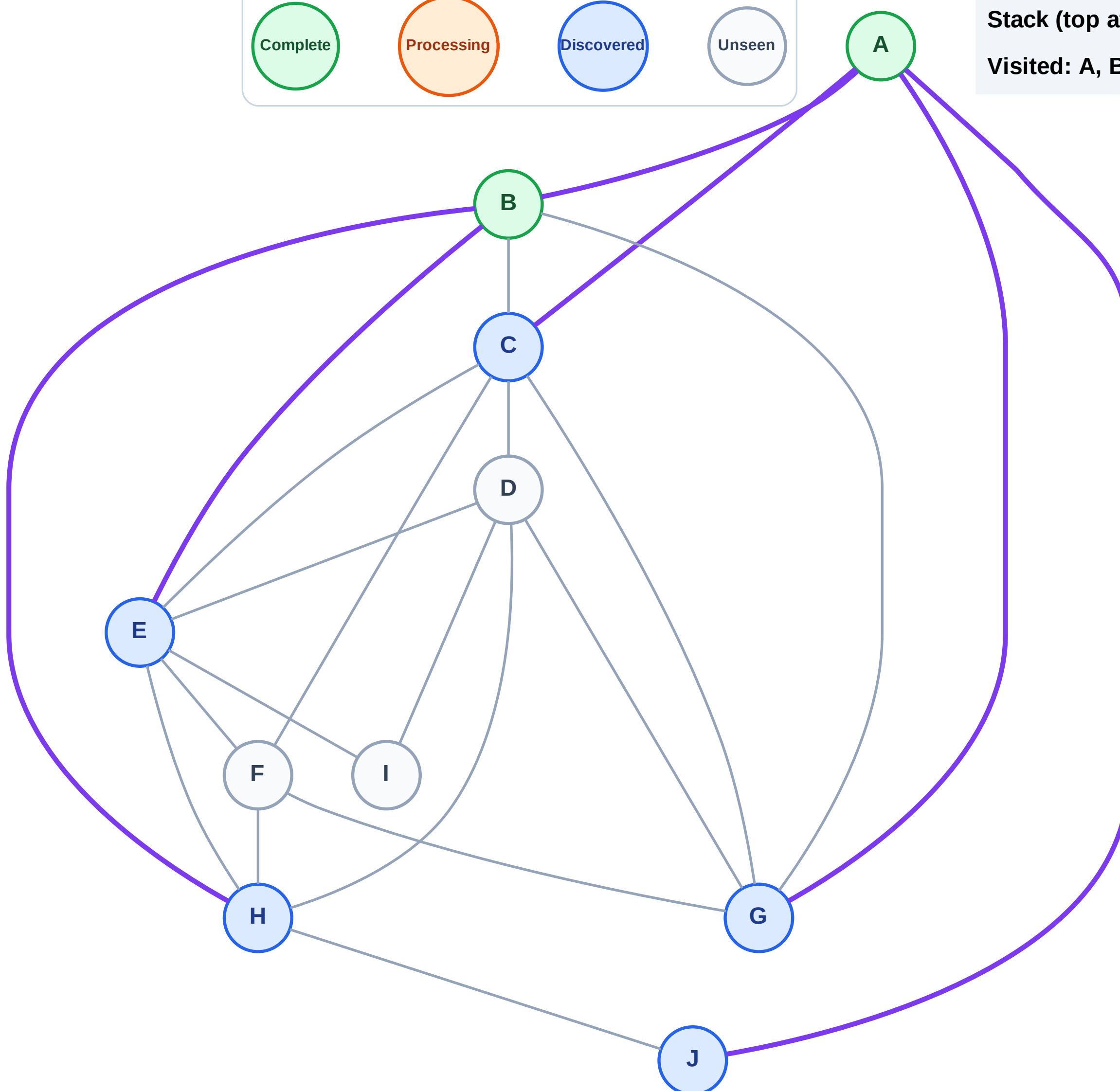
Step 5: Discover H, E from B

Legend



Stack (top at right): J | G | C | H | E

Visited: A, B



DFS Traversal

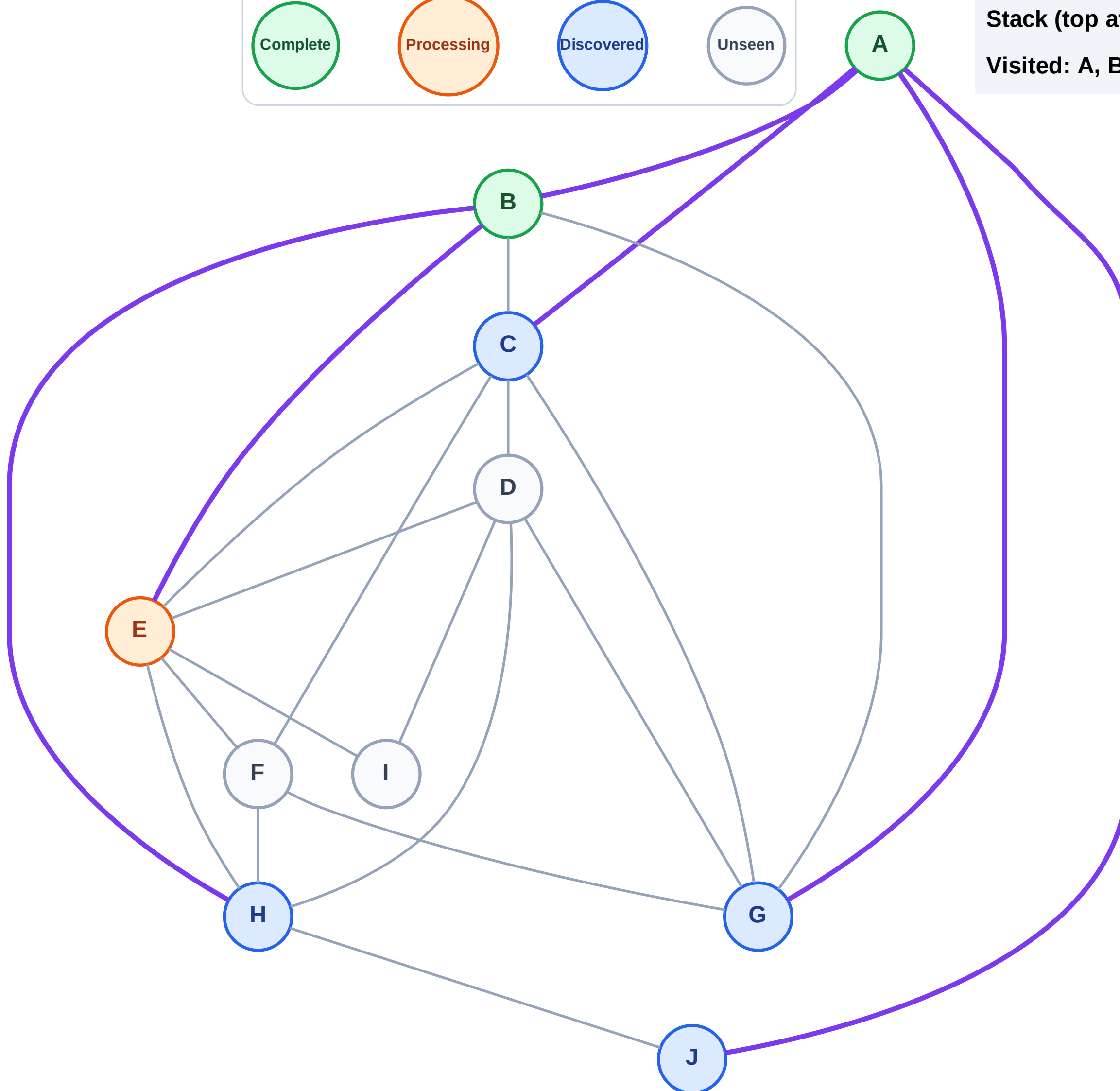
Step 6: Process node E

Legend



Stack (top at right): J | G | C | H

Visited: A, B



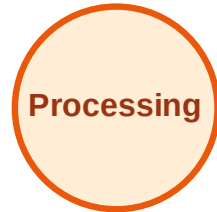
DFS Traversal

Step 7: Discover I, F, D from E

Legend



Complete



Processing



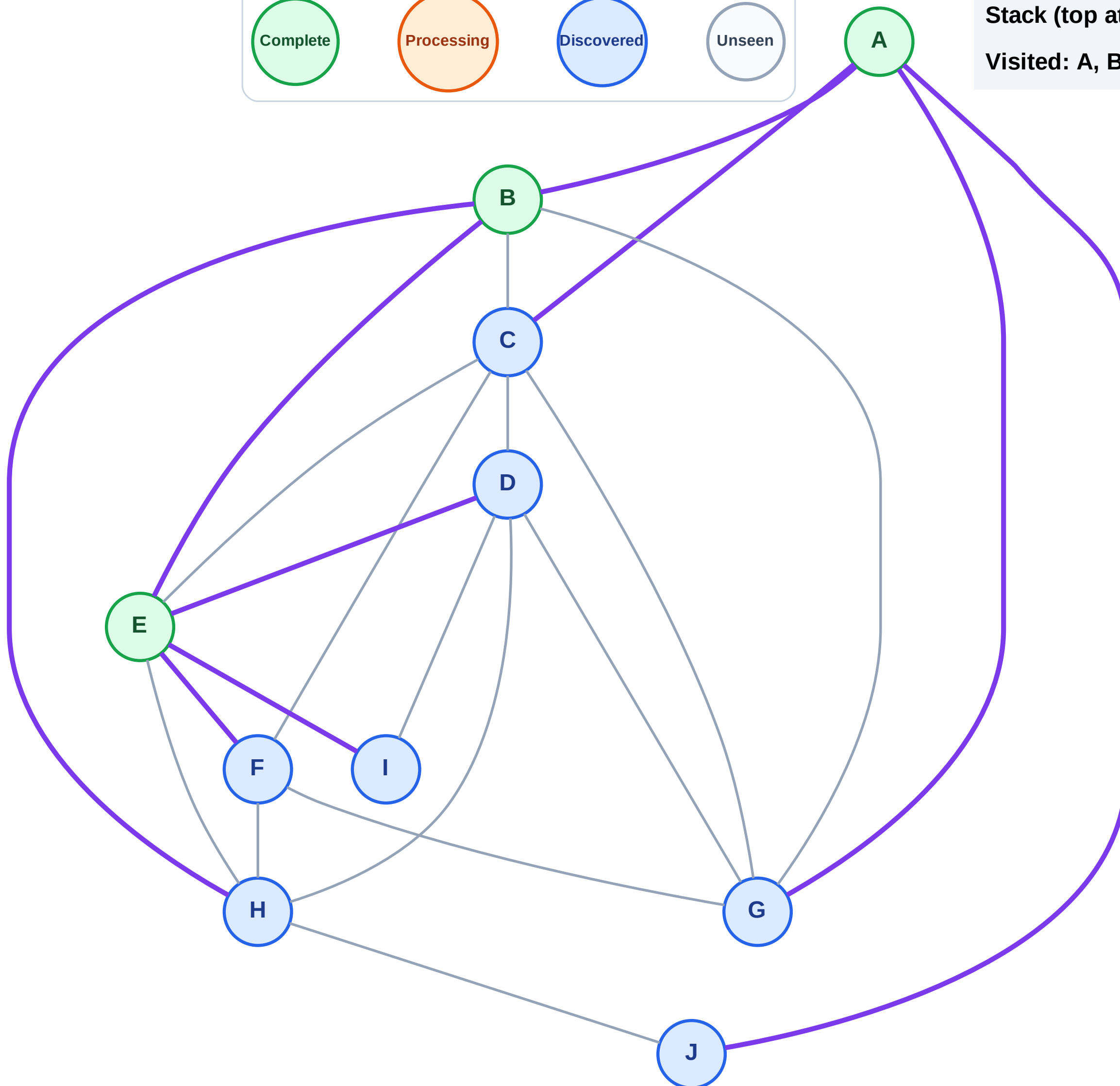
Discovered



Unseen

Stack (top at right): J | G | C | H | I | F | D

Visited: A, B, E



DFS Traversal

Step 8: Process node D

Legend

Complete

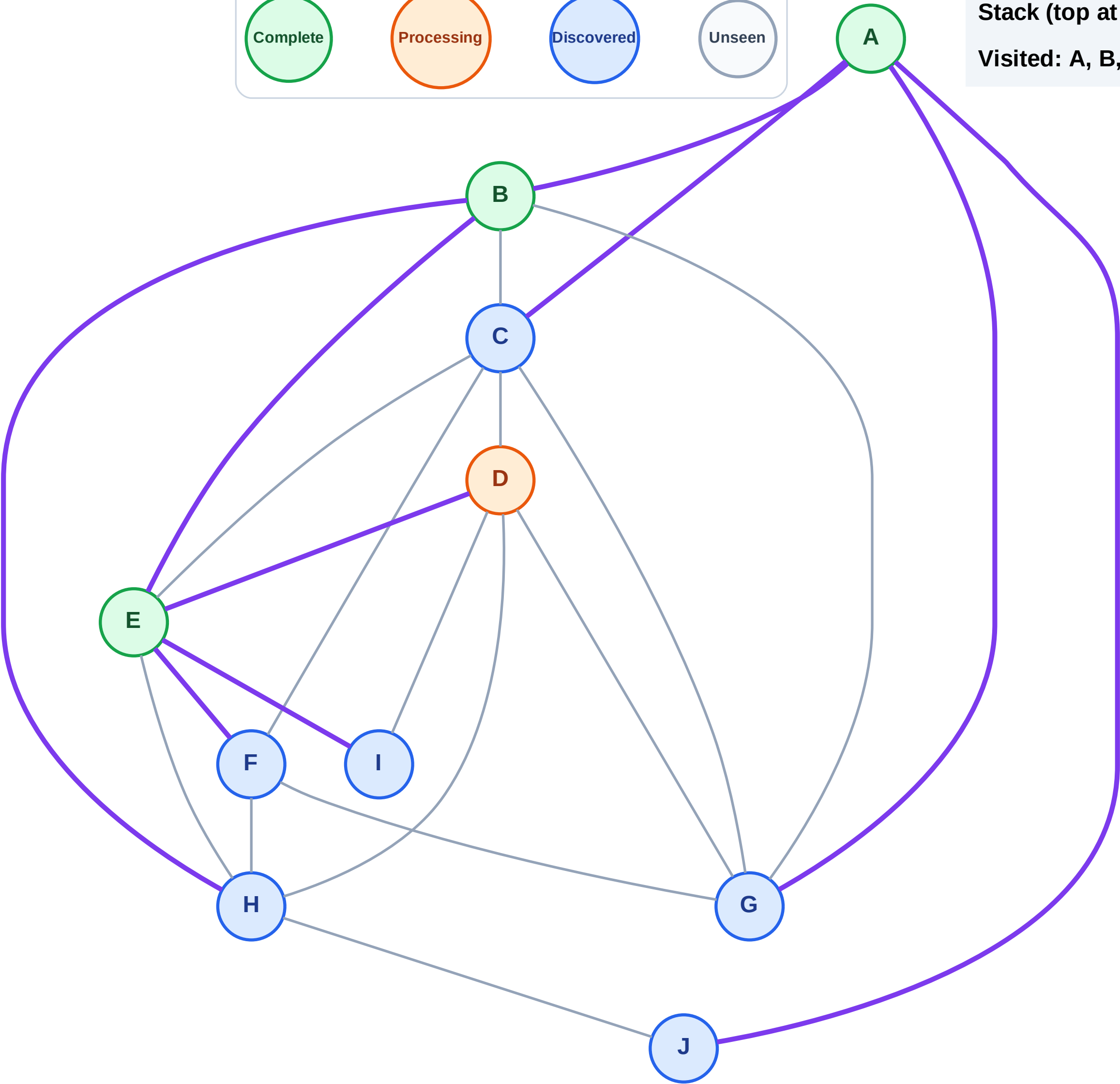
Processing

Discovered

Unseen

Stack (top at right): J | G | C | H | I | F

Visited: A, B, E



DFS Traversal

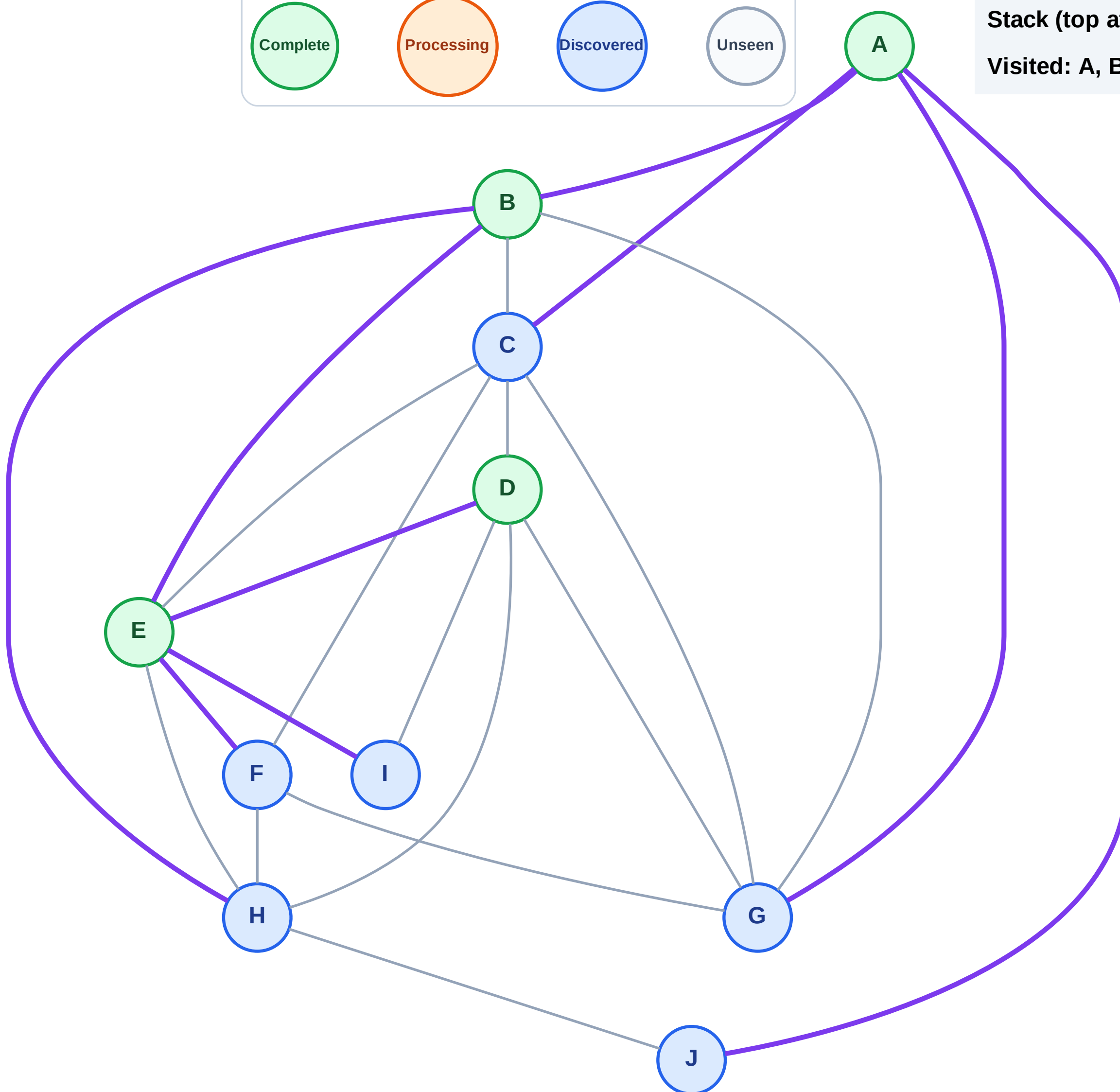
Step 9: No new neighbors from D

Legend



Stack (top at right): J | G | C | H | I | F

Visited: A, B, D, E



DFS Traversal

Step 10: Process node F

Legend

Complete

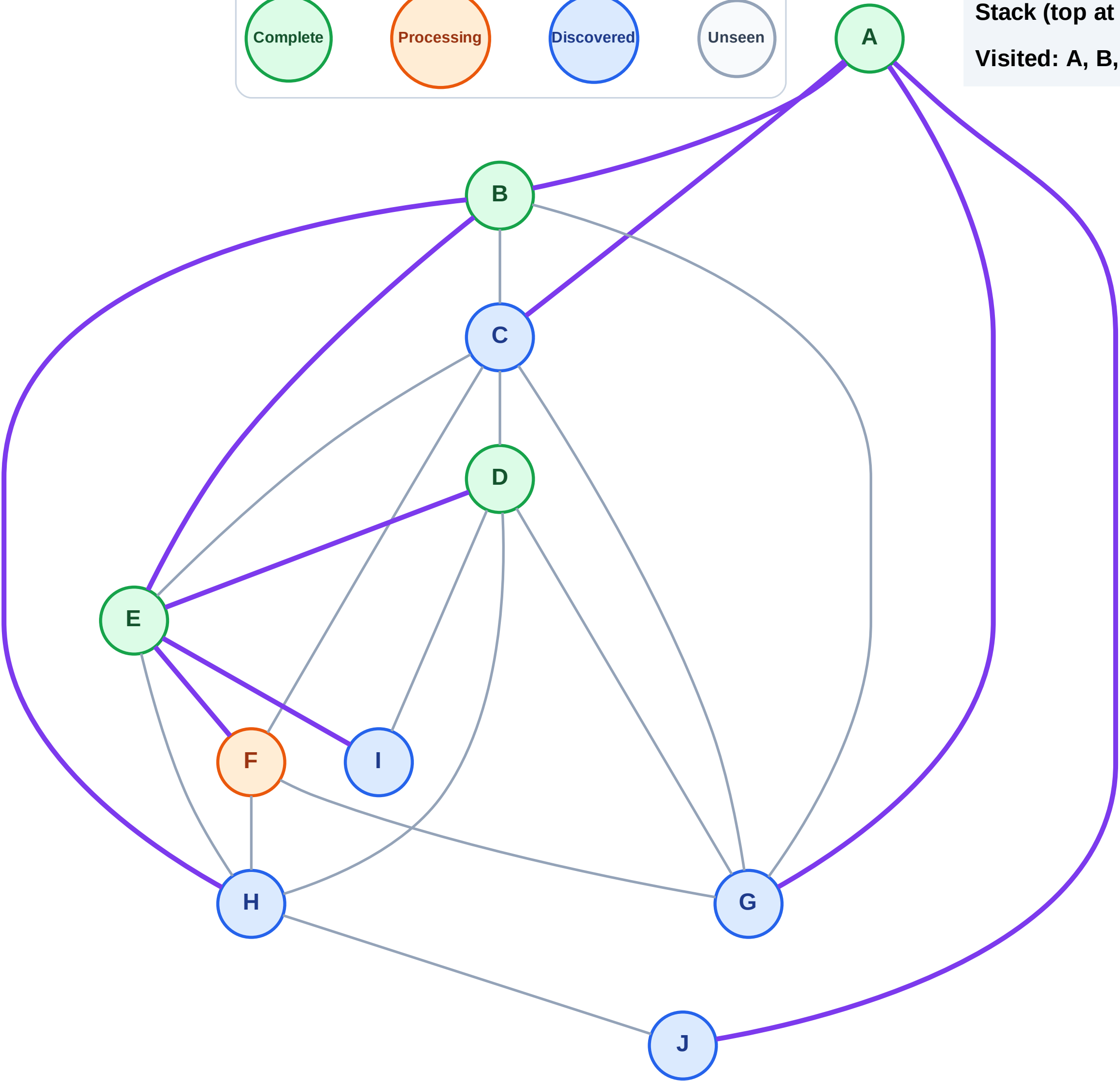
Processing

Discovered

Unseen

Stack (top at right): J | G | C | H | I

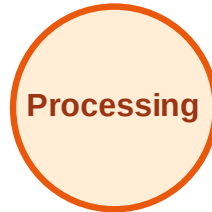
Visited: A, B, D, E



DFS Traversal

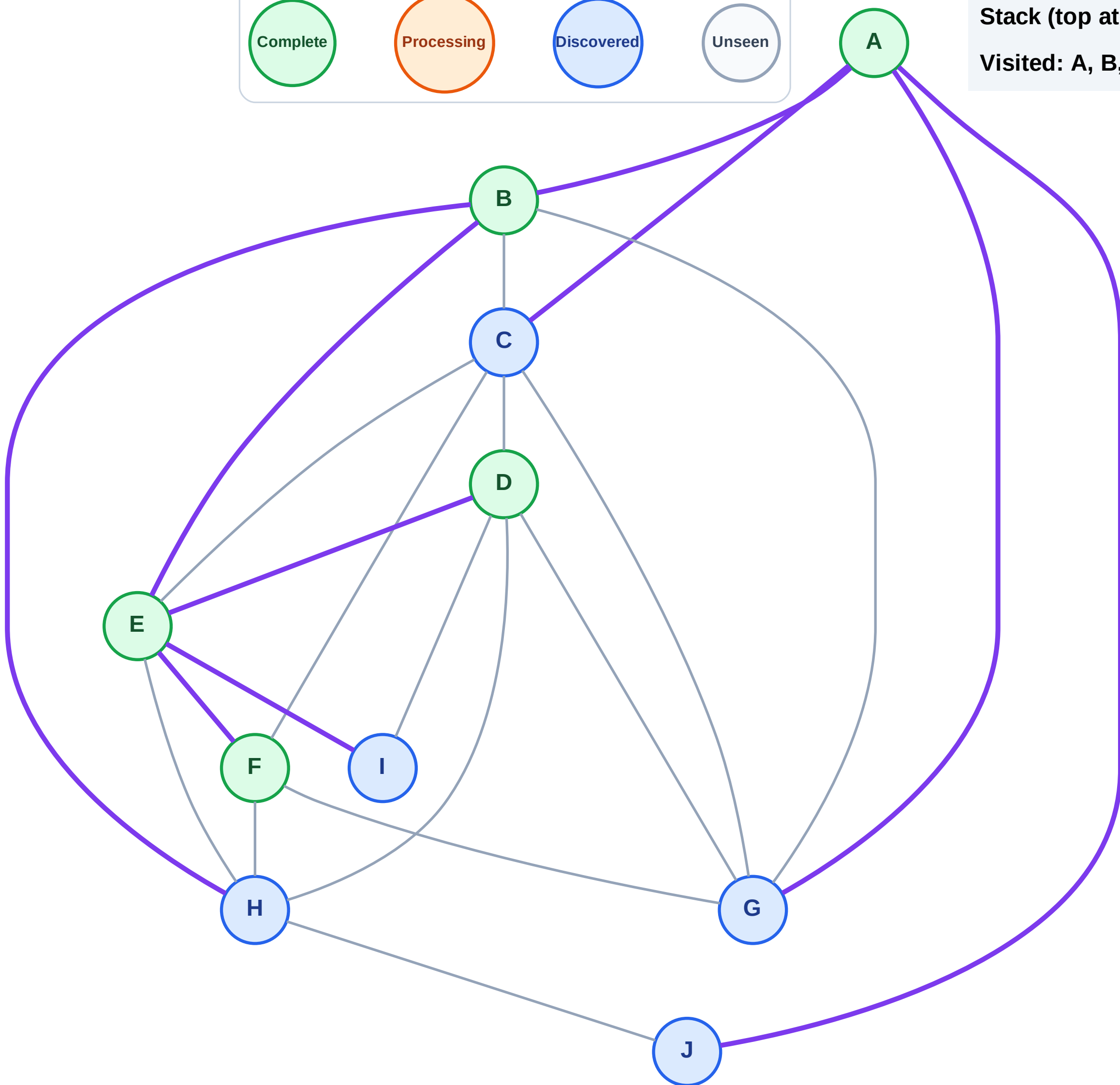
Step 11: No new neighbors from F

Legend



Stack (top at right): J | G | C | H | I

Visited: A, B, D, E, F



DFS Traversal

Step 12: Process node I

Legend

Complete

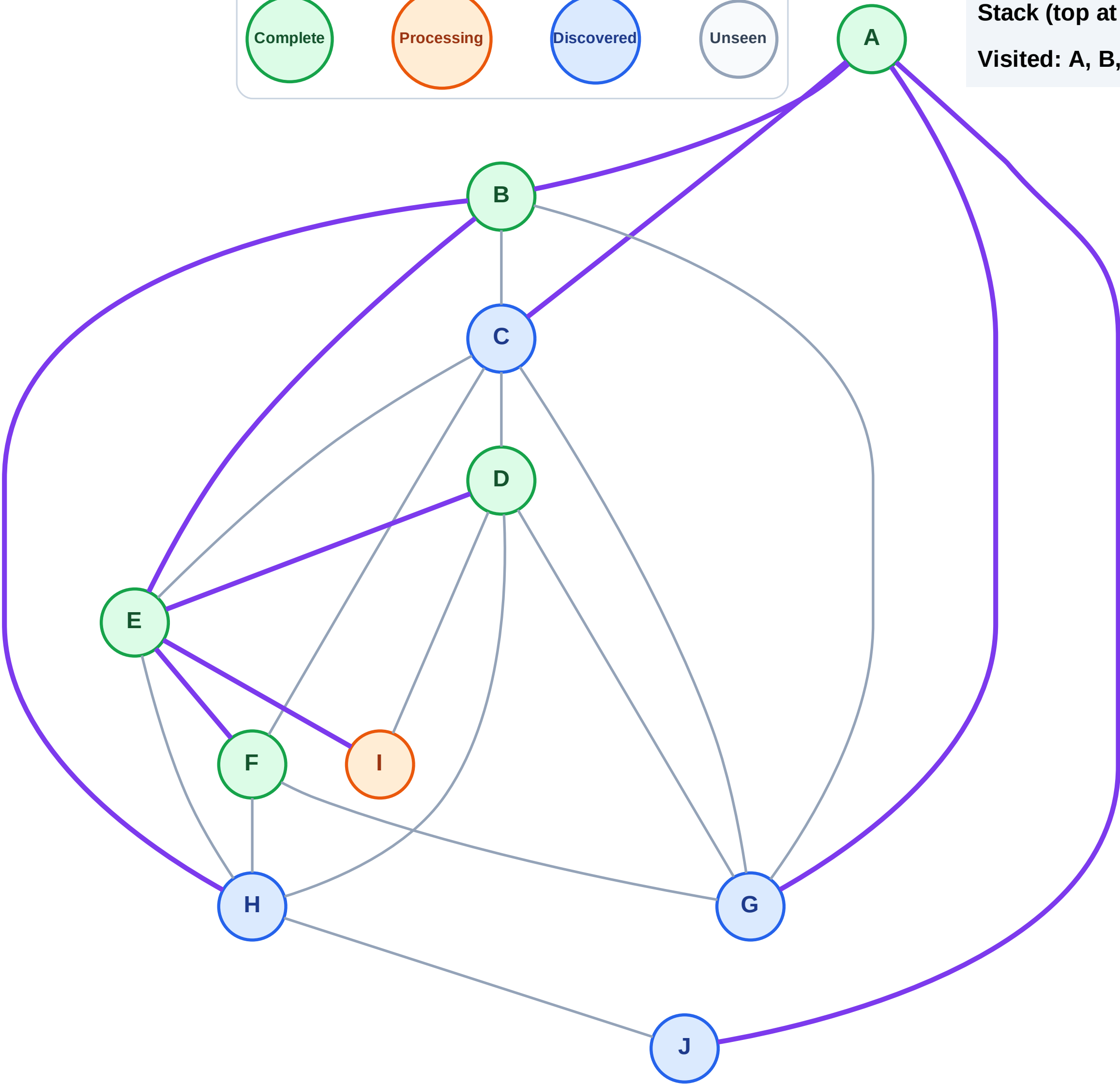
Processing

Discovered

Unseen

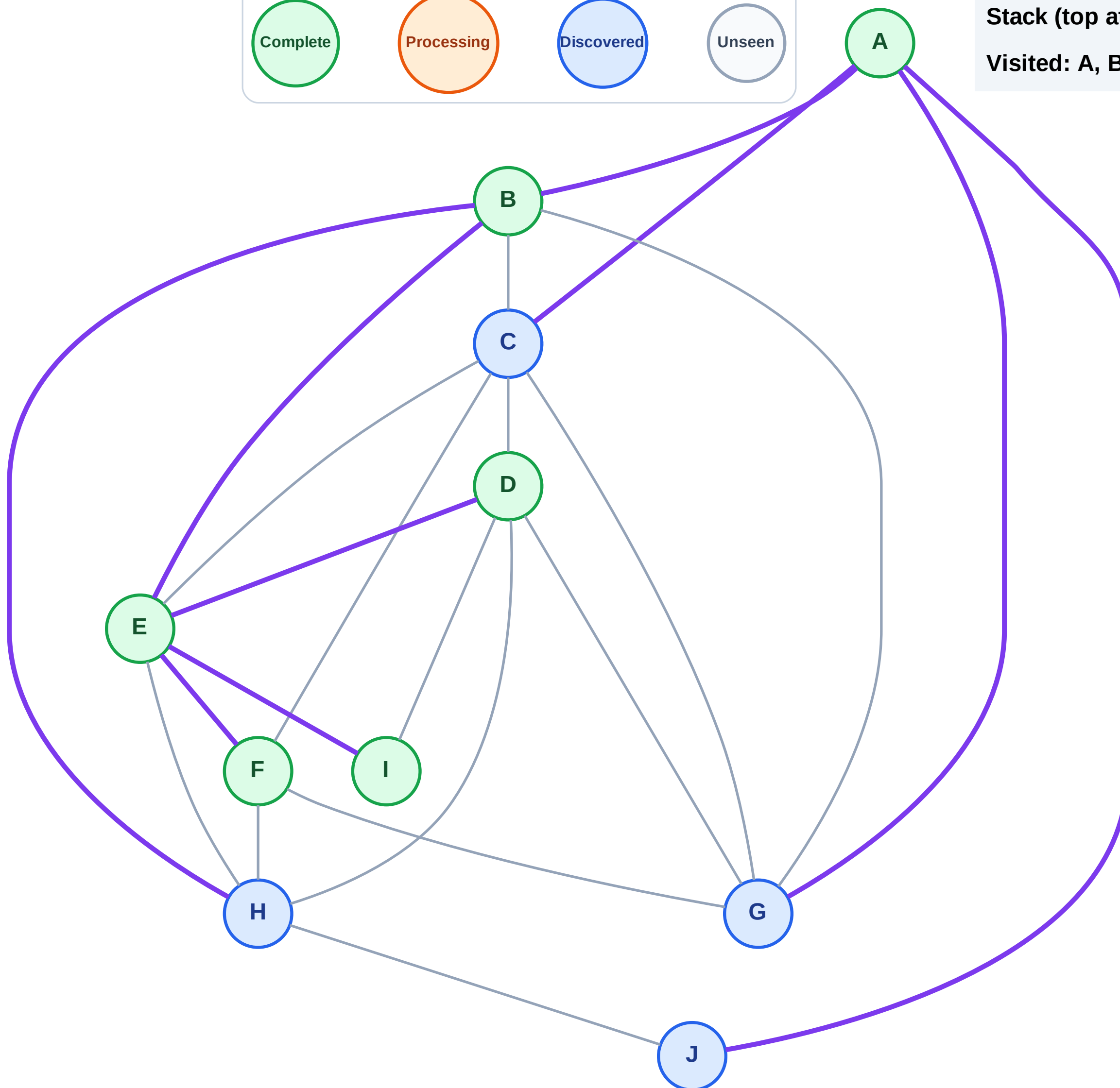
Stack (top at right): J | G | C | H

Visited: A, B, D, E, F



Step 13: No new neighbors from I

Step 13: No new neighbors from I

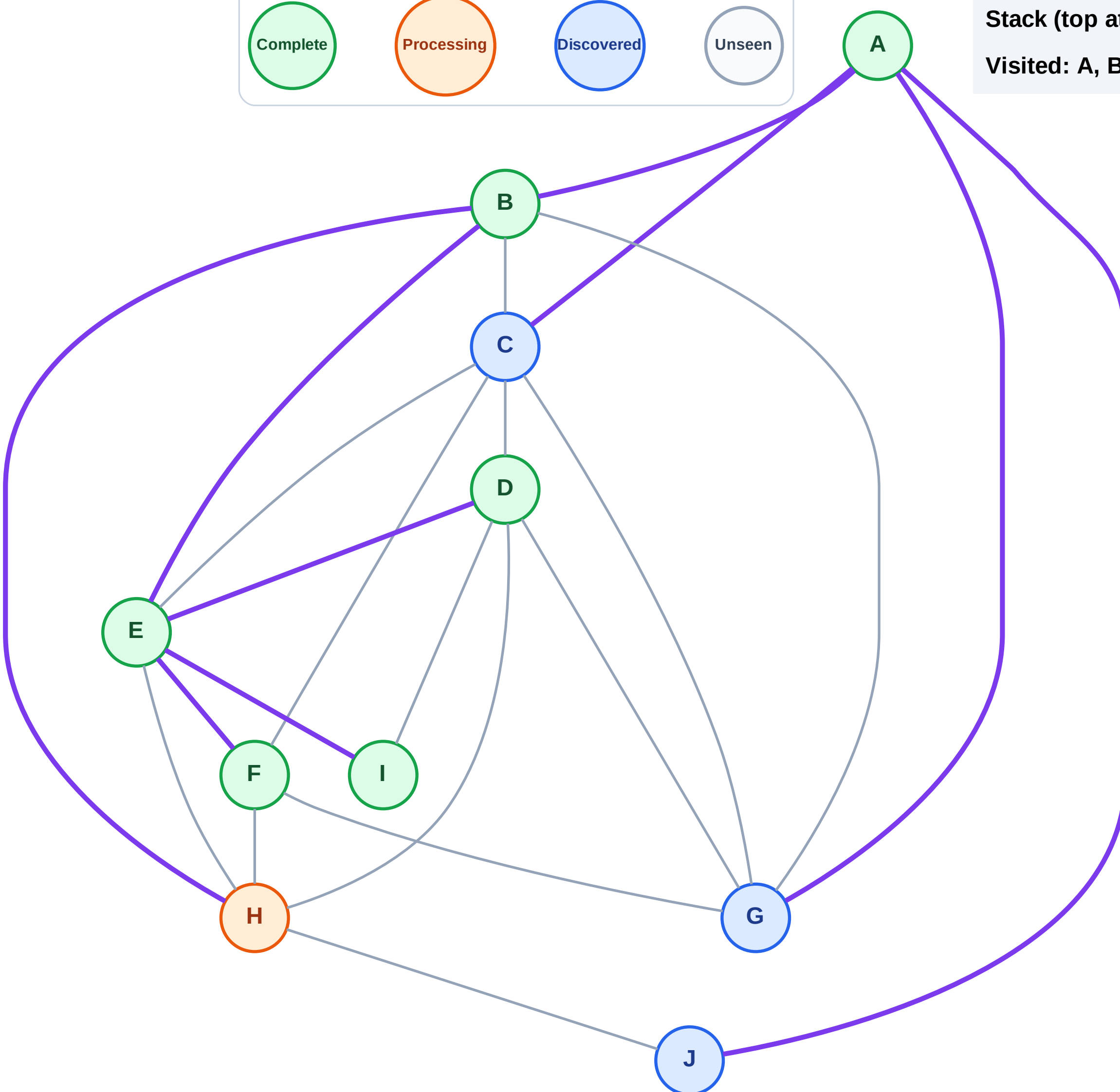


Step 14: Process node H

Legend

The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle with the label 'Complete', an orange circle with the label 'Processing', a blue circle with the label 'Discovered', and a grey circle with the label 'Unseen'.

Visited: A, B, D, E, F, I



DFS Traversal

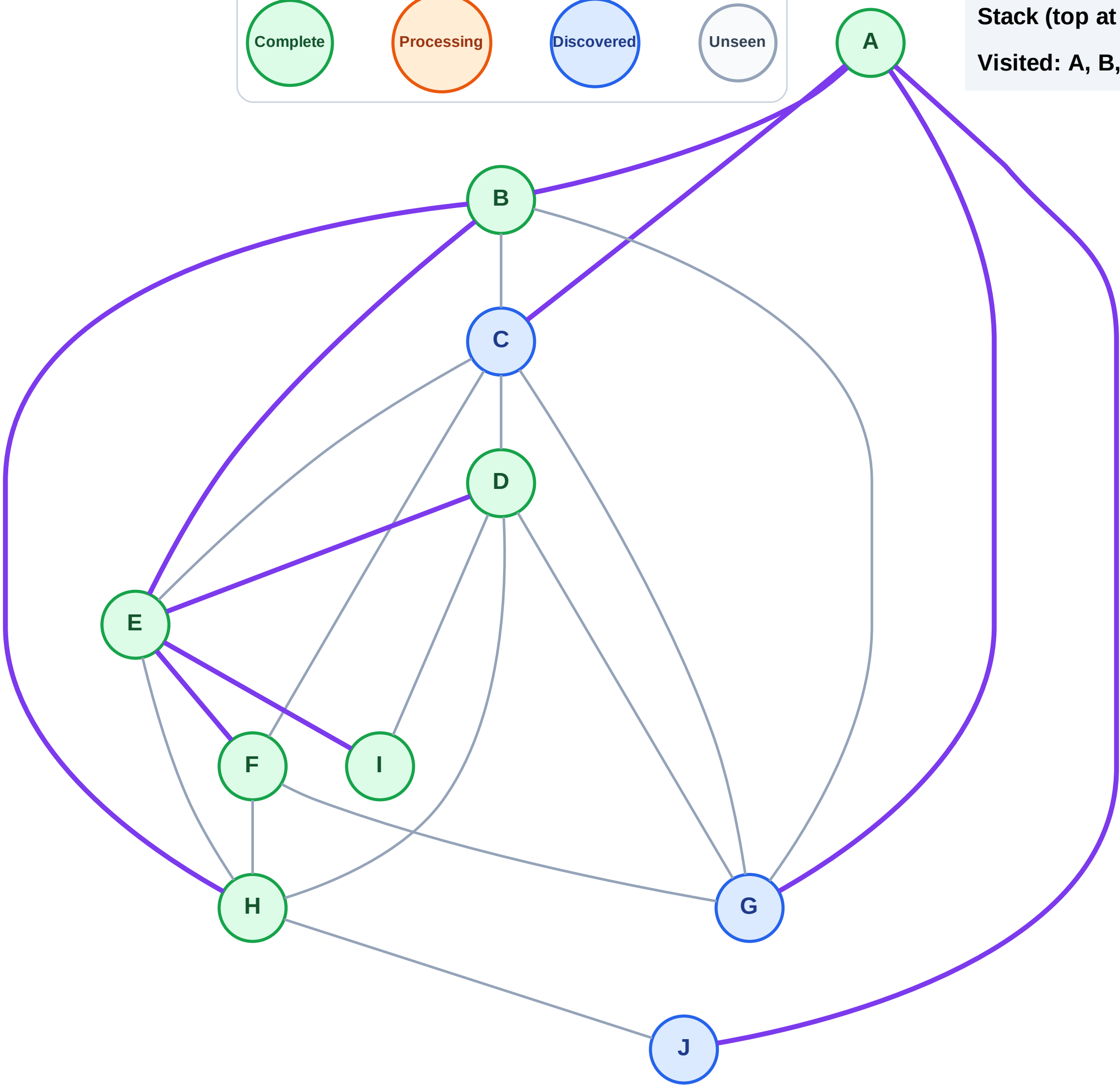
Step 15: No new neighbors from H

Legend



Stack (top at right): J | G | C

Visited: A, B, D, E, F, H, I



DFS Traversal

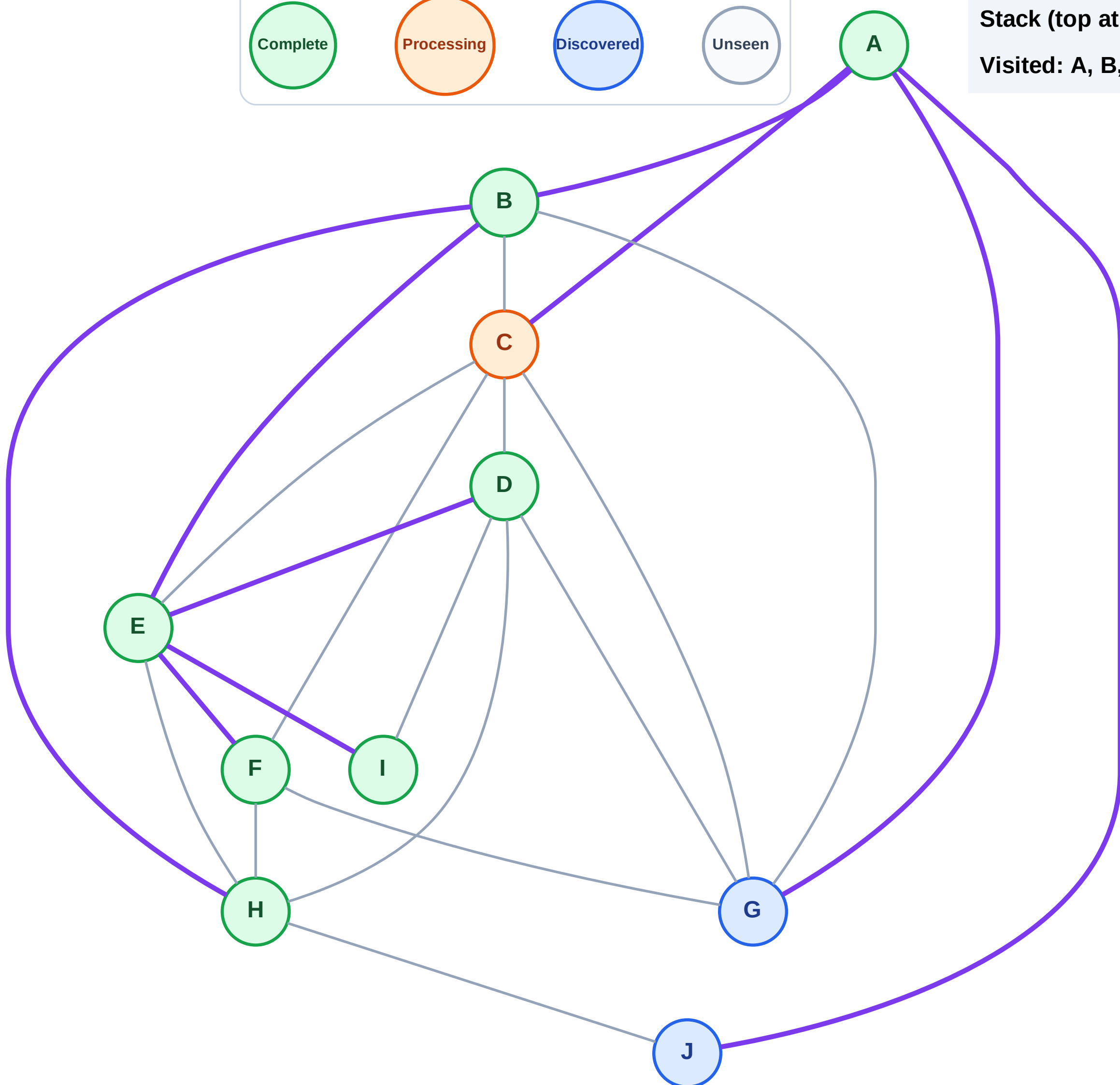
Step 16: Process node C

Legend



Stack (top at right): J | G

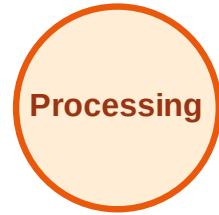
Visited: A, B, D, E, F, H, I



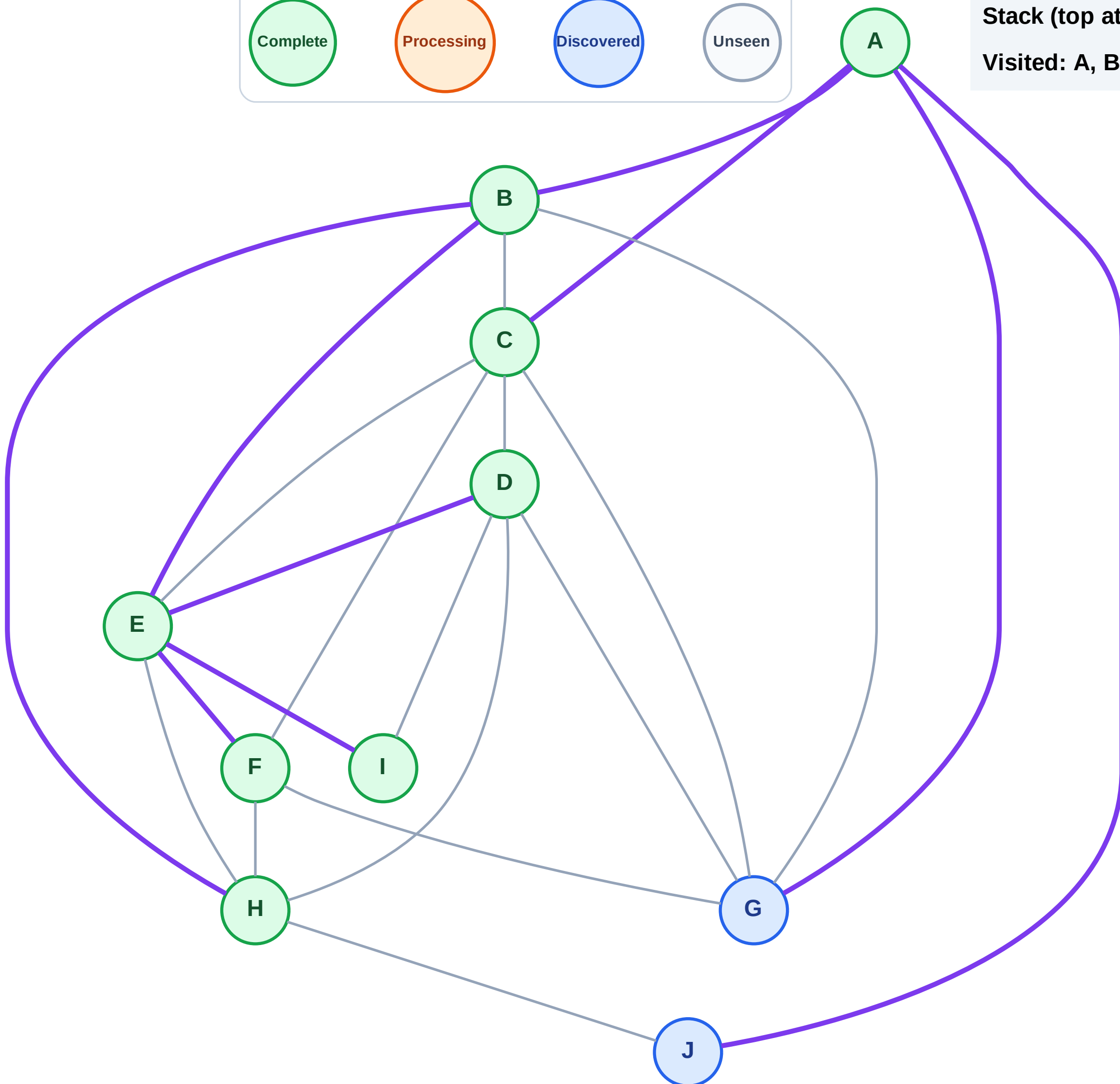
DFS Traversal

Step 17: No new neighbors from C

Legend



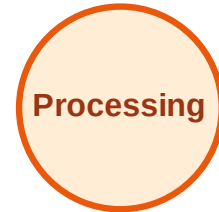
Stack (top at right): J | G
Visited: A, B, C, D, E, F, H, I



DFS Traversal

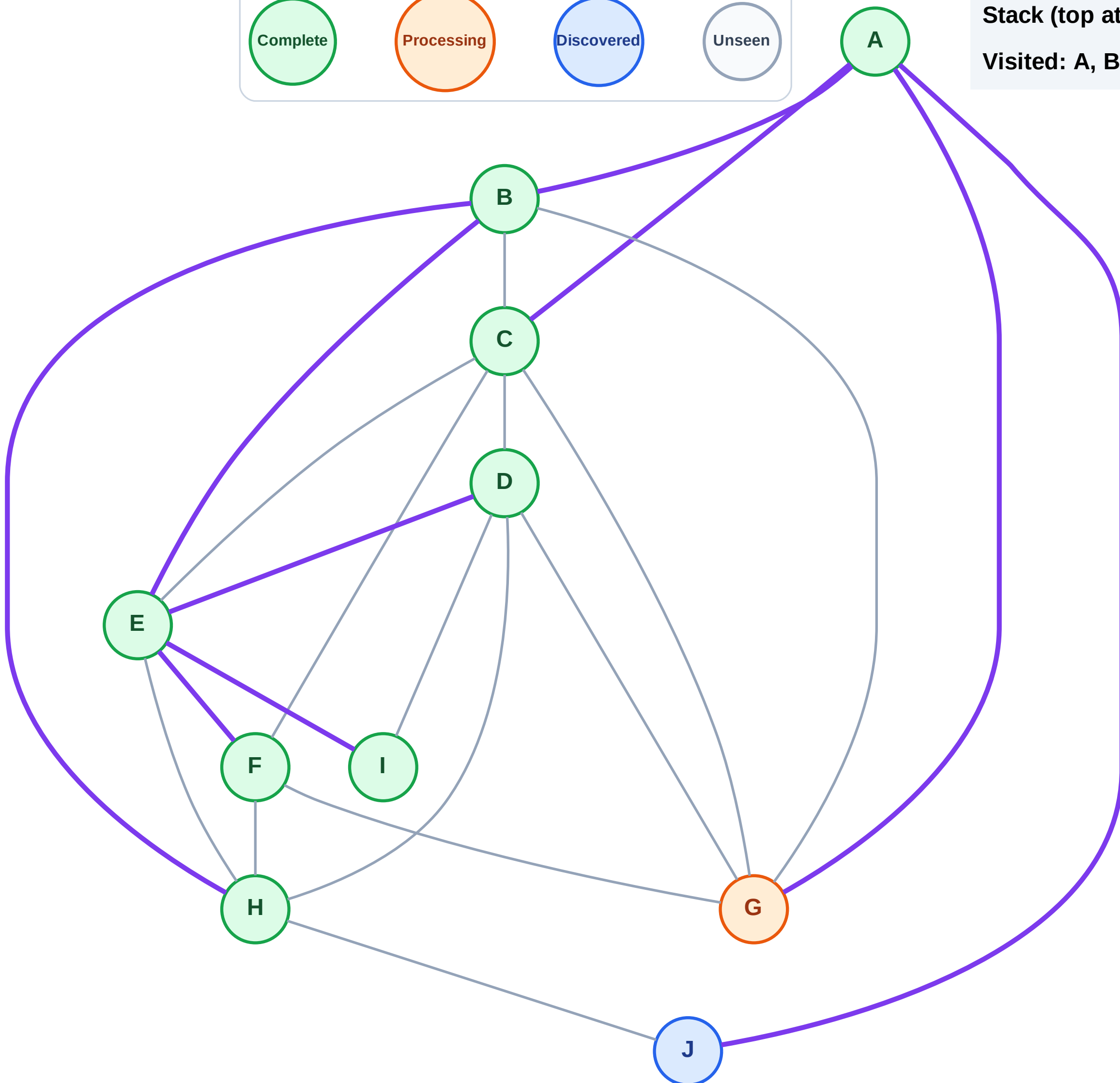
Step 18: Process node G

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, H, I



DFS Traversal

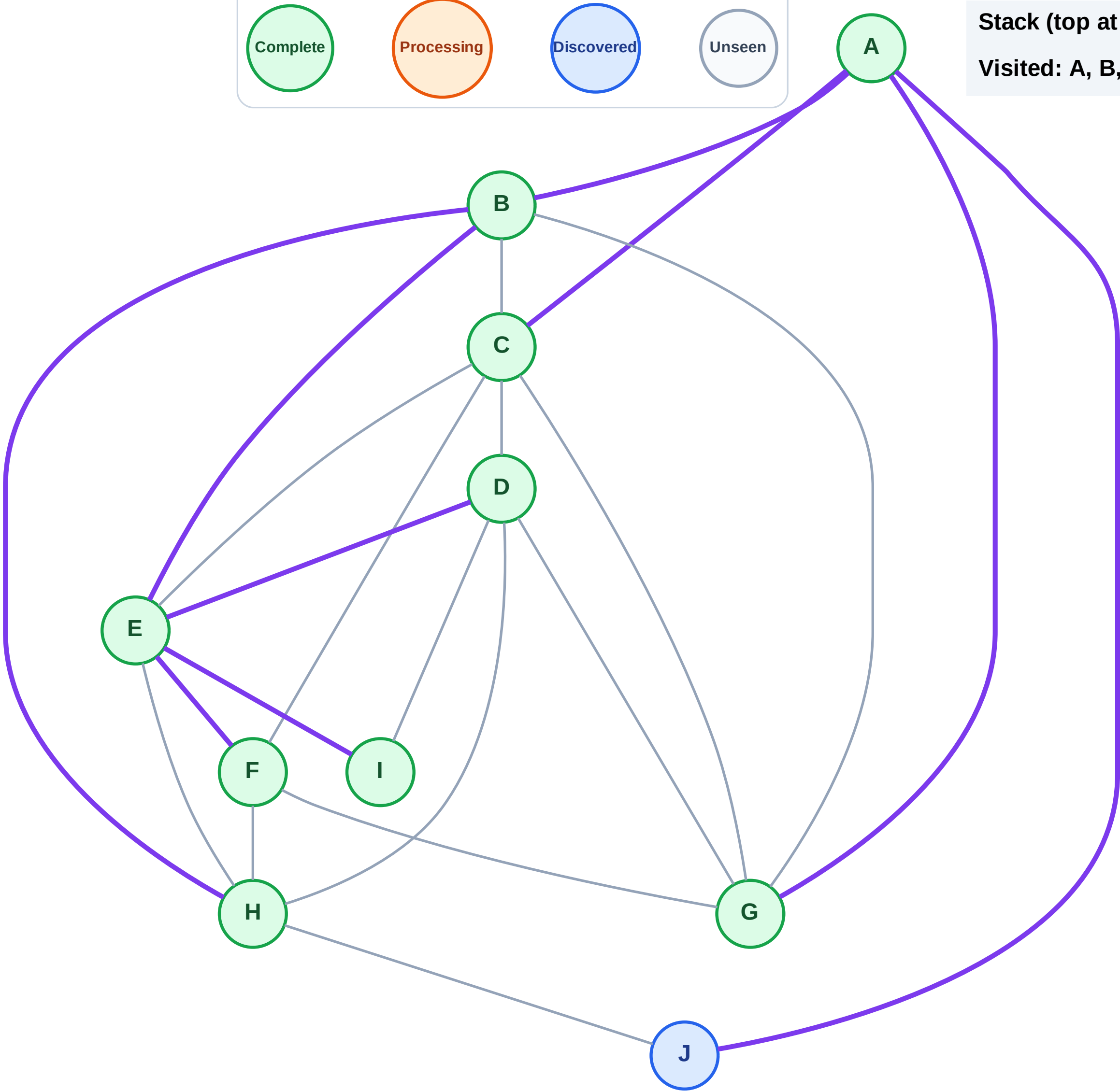
Step 19: No new neighbors from G

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

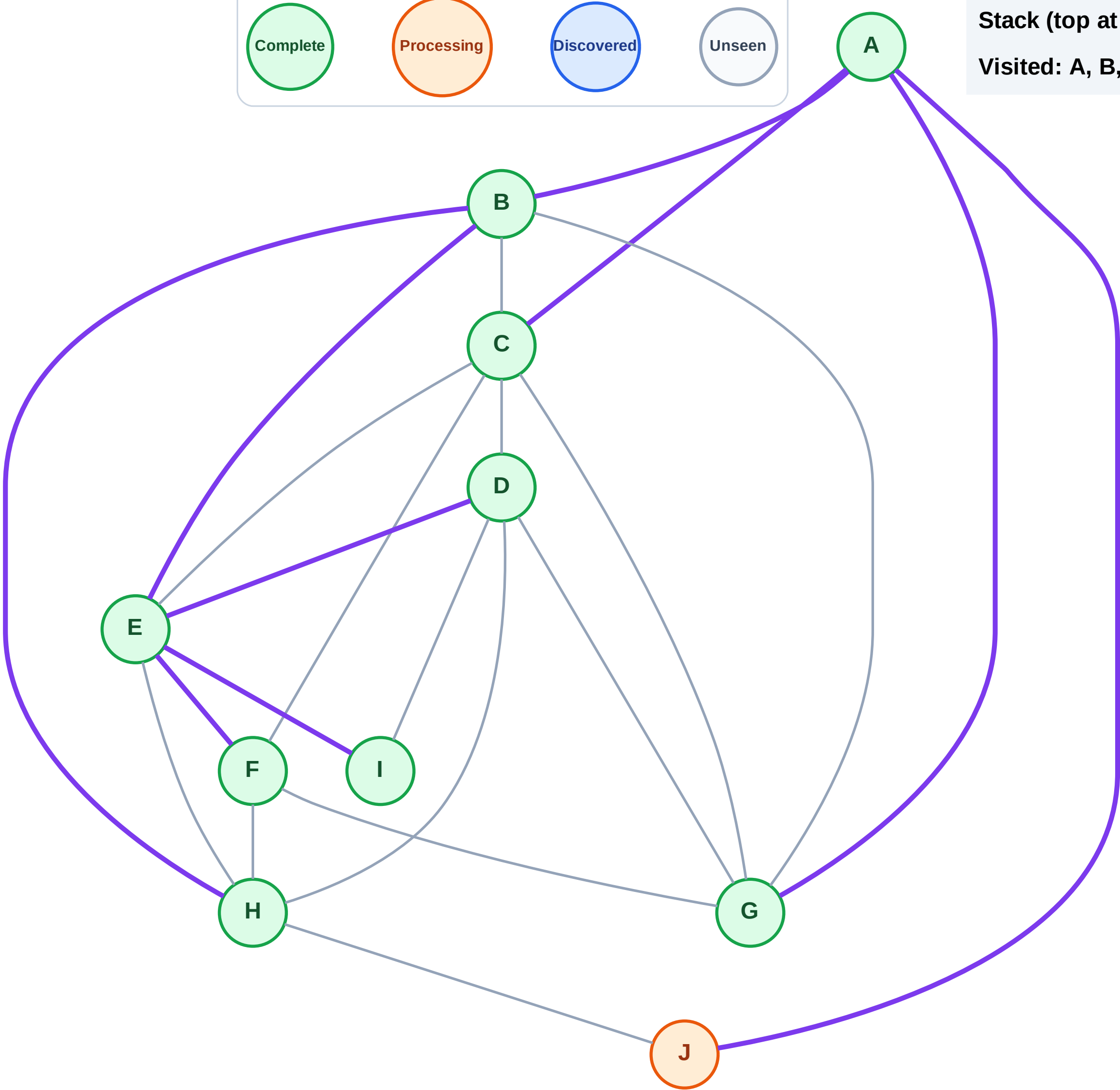
Step 20: Process node J

Legend



Stack (top at right): (empty)

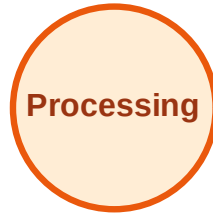
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

